

Computation of the unramified Plancherel measure

Research note prepared for the author

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Abstract

The one-line proof of Theorem 4.5.8 in the supplied manuscript is insufficient: the cited theorem assumes both a rank-one functional equation and a virtual-color condition, and its Euler factor is not defined there using the manuscript's V_X . This note supplies explicit cone and reflection certificates, together with graded Lie-centralizer calculations, for every no-type-N family in Theorem 1.1.28. The certificates prove the relevant instance of Sakellaridis's Statement 7.1.5; it is not retained as an unproved assumption. Exceptional rank-one supports are treated using the later scattering theorem and an explicit unramified quadratic-twist calculation. The arithmetic multiplicity-one hypothesis is kept separate: several literal simply connected entries fail it. The note also corrects the positive-part definition, identifies necessary parameter and central-form qualifications in the classification, and records grading discrepancies in the 2013 source.

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1 Verdict, scope, and a usable replacement statement

The formula in Theorem 4.5.8 is supported by the calculations below on the connected homogeneous affine families requested in Theorem 1.1.28, after the definition and local hypotheses are made precise. Its printed proof, “This is [Sak13, Theorem 7.2.1],” should be replaced by an argument that includes the missing combinatorial and representation comparisons. This is a proof audit of those families, including exclusions; it is not an assertion that every entry is multiplicity free over a local field.

The algebraically closed classification and the local theorem use different base fields. For geometry, pass to an algebraic closure of characteristic zero. For the local formula let k be a nonarchimedean local field of characteristic zero and odd residue characteristic, as in the manuscript’s local setup, let q be its residue cardinality, and retain the integral-model and orbit hypotheses of [Sak13, §1.7, Axioms 2.2.1, 2.4.1 and 2.4.2] used in defining its spherical functions. These hold at almost all places for a global model. Being an unramified *group* does not by itself state those hypotheses for X . No claim about an arbitrary bad-reduction model or omitted small-characteristic hypothesis is made here. The rank-one quadratic calculations use the same valid integral-model conventions as the torus calculations cited in their proof.

Theorem 1 (Verified form of the split-group assertion). *Let $X = H \backslash G$ be a compatible unramified form of one of the connected homogeneous affine families in Theorem 1.1.28, with its actual subgroup image and character lattice, or a specialization identified with another entry. Let G be split and impose the local hypotheses just specified. Exclude the roots of type N and impose the manuscript’s condition*

$$\dim \operatorname{Hom}_{G(k)}(\pi, C^\infty(X(k))) \leq 1 \quad \text{for every irreducible unramified } \pi. \quad (\text{V1})$$

Use the corrected positive part below, the grading of Definition 4.5.5, and the even-sphere Frobenius action. Then, as rational functions of the unramified parameter,

$$B_w(\chi) = \frac{\beta_X(\chi)}{\beta_X(w\chi)}, \quad \beta_X(\chi) = \frac{L(\chi, V_X^+)}{L(0, \chi, \check{\mathfrak{g}}_X^+)}, \quad w \in W_X. \quad (\text{V2})$$

An Euler factor in degree d is evaluated at $s = d/2$. Values at singular parameters are understood by meromorphic continuation. The conclusion is insensitive to multiplying β_X by a nonzero W_X -invariant rational function, including a constant zeta factor.

Proof architecture. The reflection lemma below reduces (V2) to the signed crossing multiset for each simple spherical reflection. Every requested geometric family is assigned a complete certificate in the coverage table below. The centralizer calculations identify the positive nonzero graded weights with those certificates; the degree-one positivity comparison is supplied separately. For classical spherical-root supports, [Sak13, Theorem 6.3.2] provides the rank-one analytic input, with the indicated printed grading errors corrected by the virtual-color formula and the later rank-one calculation. The two exceptional supports are checked explicitly using [Sak25, Theorem 2.4.2]; the nonsplit G_2/A_2 form is calculated by descent to its nonsplit terminal torus. The final section records the Frobenius signs in all families where they can change the nonzero Euler weights. The cocycle relation then gives (V2) for every w .

The theorem is deliberately tied to the connected classification being checked. The manuscript’s broader wording includes other stabilizers unless connectedness or a reduction is stated. A claim covering arbitrary disconnected reductive H , or arbitrary nonsplit ambient G , requires its own reduction and is not supplied by this classification check. Likewise, arbitrary ramified forms inside a split ambient group require inertia-fixed Euler factors and local calculations beyond the unramified forms treated here.

1.1 Corrections to Definitions 4.5.1–4.5.6

Fix a compatible ambient dual Borel $\check{B} = \check{T}\check{N}$ and put $\check{\mathfrak{n}}^+ = \text{Lie}(\check{N})$. Let $p : \check{\mathfrak{g}}^e \rightarrow \check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X$ be the quotient map. The precise replacement for the ambiguous Lie-algebra term in Definition 4.5.6 is

$$Q_X^+ := p(\check{\mathfrak{g}}^e \cap \check{\mathfrak{n}}^+) \simeq \frac{\check{\mathfrak{g}}^e \cap \check{\mathfrak{n}}^+}{\check{\mathfrak{g}}_X \cap \check{\mathfrak{n}}^+}, \quad V_X^+ := S_X^+ \oplus Q_X^+. \quad (\text{V3})$$

The denominator is valid because $\check{\mathfrak{g}}_X$ centralizes e . Positive *roots* of $\check{G}$ index its root spaces; positive coroots do not. The resulting positive part is a graded representation of $\check{T}_X \rtimes \Gamma$, subject to the stated Frobenius stability, and generally is not a representation of the whole $\check{G}_X$. Likewise $\check{\mathfrak{g}}_X^+$ in the denominator of (V2) is its positive-root torus representation. Definition 4.5.6 uses the positive decomposition from Conjecture 4.5.2 as well as the action from Conjecture 4.5.3; it should reference both.

The highest-weight recipe in Definition 4.5.1 does not, for an arbitrary representation, imply that its weights are just the Weyl translates of the proposed highest weights. Interior weights can occur. In the present case calculations the degree-one constituents are explicitly identified and minuscule (or tensor products of minuscule modules for different factors), so the assertion is verified there. It should not be built into a general definition without proof. Repeated colors are multiplicity labels, not distinct linearly independent elements of a lattice. If a central extension is used to separate them, distinguish the lifted construction from its descent. Also restore item (3), concerning simple root operators, when quoting [BZSV, Conjecture 4.3.16] in Conjecture 4.5.2.

Finally, “Assumption 4.5.1” currently points to the number of a definition. The multiplicity hypothesis is equation (4.5.1); reference that equation or give the hypothesis a separate number. The first sentence of §4.5 should not declare the local base field algebraically closed.

1.2 Coverage of the requested classification

The simple-group case is expanded into the complete classical ledger and the exceptional table below. This table locates every other item.

Item in 1.1.28	Family or treatment	Location
1, C1, C7	Type N except the rank-one torus endpoints	Classical ledger; C2
1, C2, C3, C5, C6	Two general-linear blocks; symplectic subgroup in odd degree	§7
1, C4	$\text{Sp}_{2n} \subset \text{SL}_{2n}$	§8
1, C8, C9	The two symplectic block families	§9
1, C10, C11, C14	General-linear and special-linear orthogonal embeddings	§10
1, C15	Spheres and rank-two even orthogonal blocks; other blocks type N	§11
1, C12, C13, C16, C17	Exceptional stabilizers in classical groups	§12
1, exceptional	All fourteen exceptional rows; eight excluded by type N	§13
2	Diagonal simple group, including exceptional type	Diagonal proposition below
3, 4(a), 4(b), 5–7	All symplectic product families and endpoints	§14
4(c), 4(d)	Orthogonal Gross–Prasad and triality	§15
8, 9	Adjacent special-linear degrees and the mixed SL/Sp family	§17

The improper pair $H = G$ is a point and has $W_X = 1$, so its cocycle identity is immediate. Any scalar Arthur-centralizer factors cancel. Trivial arms and nonsimple low-rank ambient groups are assigned to their indicated product or torus specializations, rather than counted again.

In item 9 the version without $\mathbb{G}_m$ requires $m + 2 \geq 5$. For the torus-containing version the displayed map has kernel μ_m : $u = I_2$, $w = I$, $\lambda^m = 1$, and $v = \lambda^2 I_m$. Thus H must mean its image (or the source modulo this kernel). If $m = 0$ is permitted, the torus is entirely ineffective and the reduced pair is item 4(a); the usual statement takes $m \geq 1$. Orthogonal symbols in a classification whose ambient group is required to be simply connected must be replaced by the specified Spin lifts. Connected lifts, full inverse images, normalizers and finite central enlargements are different choices and are not interchangeable.

The correct classification source used here is Bravi–Pezzini, *The spherical systems of the wonderful reductive subgroups*, J. Lie Theory **25** (2015), 105–123, especially §3. It should not be confused with their different paper on wonderful subgroups in J. Algebra **409** (2014).

2 What is sufficient to prove the cocycle formula

The statement to be proved is an identity of rational functions. Values at singular parameters are interpreted by meromorphic continuation. Write $T = \check{T}_X$, $W = W_X$, and let R^+ be the positive roots of $\check{G}_X$. For a finite multiset P of triples $a = (\lambda, \sigma, r)$, with $\lambda \in X^*(T)$, $\sigma \in \mathbf{C}^\times$, and $r > 0$, put

$$E_P(t) = \prod_{(\lambda, \sigma, r) \in P} (1 - \sigma q^{-r} \lambda(t)), \quad D(t) = \prod_{\alpha \in R^+} (1 - \alpha(t)), \quad \beta_P(t) = D(t)/E_P(t).$$

Multiplicities are retained in all products. The Weyl group acts on the first component of a triple; put $-a = (-\lambda, \sigma, r)$. This last convention retains the eigenvalue σ , as in Sakellaridis’s signed virtual colors. It is an assumption on a multiset, not an assertion about the dual of an arbitrary representation.

Lemma 1 (the exact comparison needed). Let β_1, β_2 be nonzero rational functions on T . Then

$$\frac{\beta_1(t)}{\beta_1(wt)} = \frac{\beta_2(t)}{\beta_2(wt)} \quad \text{for every } w \in W \quad \Longleftrightarrow \quad \beta_1/\beta_2 \in \mathbf{C}(T)^W.$$

Proof. On the common domain of definition, cross multiplication gives $(\beta_1/\beta_2)(t) = (\beta_1/\beta_2)(wt)$. This is an identity in the rational-function field, so poles and zeros do not affect the argument. The converse is the same computation. $\square$

Consequently the equality of the full graded representations V_X and the auxiliary representation V'_X in BZSV is sufficient but is not necessary for Theorem 4.5.8. What must be checked is

$$\frac{L(t, V_X^+)}{L(t, (V'_X)^+)} \in \mathbf{C}(T)^W,$$

with compatible conventions for both positive parts and Frobenius. A difference of full representations alone does not disprove the desired cocycle formula.

Lemma 2 (reflection crossings). Suppose a finite multiset M is W -stable and has a decomposition $M = P \sqcup (-P)$, where all weights are nonzero and, for each weight λ , at most one of $\lambda, -\lambda$ occurs in P . For a simple reflection $s = s_\alpha$, let $A_s = P \cap (-sP)$, with multiset intersection. Then

$$\frac{\beta_P(t)}{\beta_P(st)} = -\alpha(t) \frac{E_{-A_s}(t)}{E_{A_s}(t)}. \tag{GL1}$$

Proof. Put $C = P \cap sP$. For each multiplicity label, $P = C \sqcup A_s$ and $sP = C \sqcup (-A_s)$. These identities can equivalently be checked by subtracting multiplicity functions; they also show $sA_s = -A_s$. Thus $E_P(st)/E_P(t) = E_{-A_s}(t)/E_{A_s}(t)$. A simple reflection permutes $R^+ \setminus \{\alpha\}$, while $(1 - \alpha(t))/(1 - \alpha(t)^{-1}) = -\alpha(t)$. Combining these identities proves (GL1). $\square$

The sign-separation hypothesis is essential. It holds when positivity is defined by a pointed cone that contains exactly one of $\lambda, -\lambda$ for every occurring weight. Zero weights can be removed first: their Euler factors are independent of t and cancel from every cocycle. A decomposition of multiplicity labels that puts both a weight and its negative in P is not sufficient for the stated crossing description.

Corollary 3 (a finite, rigorous certificate). Let Θ be Sakellaridis's signed weighted multiset, C the cone generated by color valuations, and $\mathcal{D}_\gamma$ the multiset of virtual weighted colors belonging to a simple spherical root γ . The following data prove Statement 7.1.5:

1. a linear functional ℓ strictly positive on every nonzero generator of C ;
2. for each $\theta \in \Theta$, an explicit nonnegative rational linear combination of color valuations equal to exactly one of θ and $-\theta$, and no zero weights;
3. for each simple reflection s_γ , a verified multiset identity

$$\{(\theta, \sigma, r) \in \Theta : \ell(\theta) > 0, \ell(s_\gamma \theta) < 0\} = \mathcal{D}_\gamma.$$

Proof. Strict positivity makes C pointed. The certificates in (2) imply that the sign of $\ell(\theta)$ agrees with membership in C or $-C$. Therefore (2) is the first assertion of Statement 7.1.5 and (3) is precisely its second assertion. $\square$

This certificate must include the grading and the sign, not only the underlying weights. It proves the statement for an infinite family only when the listed formulas and identities hold for the entire parameter range, including its small-rank endpoints. Numerical tests at selected ranks do not establish the family.

Corollary 4 (cocycle reduction without a global identification). Suppose the local functional-equation coefficients satisfy the cocycle relation and, for every simple spherical root γ , the rank-one calculation gives

$$B_{s_\gamma}(t) = -\check{\gamma}(t) \frac{E_{-\mathcal{D}_\gamma}(t)}{E_{\mathcal{D}_\gamma}(t)}.$$

If the positive graded Frobenius-weight multiset of V_X^+ has crossings $\mathcal{D}_\gamma$, then

$$B_w(t) = \frac{L(t, V_X^+)/L(0, t, \check{\mathfrak{g}}_X^+)}{L(wt, V_X^+)/L(0, wt, \check{\mathfrak{g}}_X^+)} \quad (w \in W_X).$$

Proof. Lemma 2 proves the equality for simple generators. A reduced expression and the cocycle relation extend it to every element of W_X . The factors telescope; independence of the expression also follows from the already given cocycle relation. $\square$

This is a conditional reduction with completely specified inputs. Neither the rank-one formula in exceptional support nor the crossing identity is proved merely by writing this reduction.

Lemma 5 (discarding a harmless extra summand). Let $U = U_0 \oplus Q$ be a graded T -representation, compatible with Frobenius F , and suppose the chosen positive part decomposes as $U^+ = U_0^+ \oplus Q^+$. If Q^+ is stable under representatives of W that commute with F , then

$$L(t, U^+)/L(t, U_0^+) = L(t, Q^+) \in \mathbf{C}(T)^W.$$

Proof. For a representative n_w , the operators $tFq^{-d/2}$ and $(wt)Fq^{-d/2}$ on each graded component of Q^+ are conjugate by n_w . Their characteristic determinants are equal. Taking the product over degrees and reciprocals proves the claim. $\square$

This gives a possible way to avoid proving $V_X = V'_X$. In applying it to the Lie-algebra summand one must actually prove the asserted splitting and stability in the nilpotent centralizer quotient. The elementary fact that a Levi subgroup preserves the positive root spaces outside its own root system is not, by itself, a proof of such a splitting in $\mathfrak{g}^e/\mathfrak{g}_X$.

3 What the cited hypotheses do and do not establish

The two independent hypotheses in Sakellaridis. Theorem 7.2.1 of *Spherical functions on spherical varieties* assumes Statements 6.3.1 and 7.1.5. Theorem 6.3.2 proves the former under the additional condition that the Levi support of every spherical root is classical. The paragraph following Statement 7.1.5 explicitly proposes its validity for affine homogeneous spaces as a conjecture. Thus a case-by-case proof needs both a rank-one analytic calculation and a global combinatorial certificate. They are logically distinct tasks.

The BZSV comparison. BZSV, Proposition 9.3.3, compares the graded torus representation obtained from the known Plancherel calculation with V_X . Its degree-one part S'_X is contained in S_X , with equality when the specified weights are minuscule. Its remaining part is described as a particular Weyl-stable subspace of the nilpotent-centralizer summand. Remark 9.3.4 states the expected general equality $V'_X = V_X$; it is not presented as a theorem. The weaker comparison in Lemma 1 is enough for the present cocycle problem, but also requires proof.

In addition, the manuscript's Conjecture 4.5.2 omits item (3) of BZSV Conjecture 4.3.16. That item controls the negative simple-root operators on the proposed positive part. Even the full statement should not be silently promoted to a proved crossing identity: a root operator records adjacent transitions in a root string, whereas a reflection reverses the whole string. As a purely representation-theoretic illustration, in $\text{Sym}^3(\mathbf{C}^2)$ the positive weights are 3, 1, and the lowering operator crosses from positive to negative only at $1 \mapsto -1$, while the reflection also sends $3 \mapsto -3$. This illustration is not a spherical-variety counterexample and is not claimed to satisfy the full BZSV construction.

4 Rational forms and the multiplicity-one restriction

Split G does not imply split color data. Even when G is split, the k -form of $H \backslash G$ can have a nontrivial Galois action on the colors. Thus the signs σ_θ in the Euler factors must be retained. For example, over a nonarchimedean field with an unramified quadratic extension E/k , both the split torus and $E^\times/k^\times$ embed as maximal tori in split PGL_2 . The two geometric colors of their quotients are fixed individually in the split case and exchanged in the nonsplit case. A two-dimensional Frobenius permutation space has eigenvalues $+1, -1$, so its factor is

$$(1 - q^{-r}z)^{-1}(1 + q^{-r}z)^{-1} = (1 - q^{-2r}z^2)^{-1},$$

whereas trivial Frobenius gives $(1 - q^{-r}z)^{-2}$. Consequently a check over the algebraic closure is not yet a check of all rational forms. Sakellaridis's split-group theorem itself retains such signs; BZSV Remark 9.3.1 explicitly notes the separate comparison needed in the nonsplit even-sphere cases.

Proposition 6 (a classification row need not meet Assumption 4.5.1). Let k be a nonarchimedean local field of characteristic zero, $G = \text{SL}_2(k)$, and $T = \{\text{diag}(a, a^{-1}) : a \in k^\times\}$. There exist irreducible unramified principal-series representations π such that

$$\dim \text{Hom}_T(\pi, \mathbf{C}) \geq |k^\times / (k^\times)^2| > 1.$$

Thus the split pair (SL_2, T) , although geometrically homogeneous affine spherical with connected reductive stabilizer, is outside the multiplicity-one hypothesis of Theorem 4.5.8.

Proof. Let B be the upper triangular Borel and let $I(s)$ be normalized smooth induction of $\text{diag}(a, a^{-1}) \mapsto |a|^s$. Thus an inducing function satisfies $f(bg) = |a|^{s+1}f(g)$ for a diagonal entry a of b . Put

$$n^-(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, \quad \phi(x) = f(n^-(x)).$$

Right translation by $t(a) = \text{diag}(a, a^{-1})$ acts by

$$(\pi_s(t(a))\phi)(x) = |a|^{s+1}\phi(a^2x).$$

For each square class $c \in k^\times / (k^\times)^2$, define

$$\ell_c(f) = \int_{c(k^\times)^2} \phi(x) |x|^{(s+1)/2} d^\times x.$$

These integrals converge when $\text{Re}(s) > -1$. Near zero, ϕ is bounded. For large $|x|$, the matrix identity

$$n^-(x) = \begin{pmatrix} x^{-1} & 1 \\ 0 & x \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}$$

shows that $\phi(x) = |x|^{-s-1}$ times a bounded locally constant function. Both ends therefore give convergent geometric series when $\text{Re}(s) > -1$. The substitution $y = a^2x$ proves $\ell_c(\pi_s(t(a))f) = \ell_c(f)$. The functionals are linearly independent: compactly supported locally constant functions in any one square class, supported away from zero, extend by zero to smooth inducing sections and are killed by all other $\ell_{c'}$. Choose s in this half-plane outside the exceptional reducibility set for principal series, so that $I(s)$ is irreducible. It is unramified by construction. Frobenius reciprocity gives the same lower bound for $\text{Hom}_G(I(s), C^\infty(T \backslash G))$. Finally $k^\times / (k^\times)^2$ has more than one element, since an element of odd valuation is not a square. $\square$

This proposition is a counterexample to applying the theorem to every geometric classification row without checking its hypotheses. It is not a counterexample to Theorem 4.5.8 with Assumption 4.5.1 retained. Passing to central covers or isogenous forms changes rational-point orbit data and can change multiplicity, so that operation must also be justified in a local statement.

Sources for this section. The primary texts used are [Sakellaridis, arXiv:0905.4244v3](#), especially Statements 6.3.1 and 7.1.5, Theorems 6.3.2 and 7.2.1, and [Ben-Zvi–Sakellaridis–Venkatesh, *Relative Langlands duality*](#), especially Conjecture 4.3.16, Proposition 9.3.3, and Remarks 9.3.1 and 9.3.4.

5 A complete uniform family: the diagonal group case

Proposition 7 (Brion case 2). Let H be a split connected reductive group over k , let $G = H \times H$, and put $X = \text{diag}(H) \backslash G$. With the compatible positive system used in the group spherical-function formula, the cocycle assertion of Theorem 4.5.8 holds, in every root type. In particular, exceptional root types of H do not create exceptional spherical-root supports in this family.

Proof. Identify X with H by $(h_1, h_2) \mapsto h_1^{-1}h_2$. The standard geometric data of the group case have the opposition-twisted diagonal dual map. Choose its pinning-compatible representative (the Chevalley involution followed by a longest-Weyl-element representative); this preserves the positive Borel and acts on the Dynkin diagram by opposition. Identify the second factor through that automorphism. In these coordinates the map is $\check{G}_X = \text{diag}(\check{H})$, with trivial Arthur SL_2 and no colors of even sphere type. Consequently $S_X = 0$, $e = 0$, and

$$V_X = (\check{\mathfrak{h}} \oplus \check{\mathfrak{h}}) / \text{diag}(\check{\mathfrak{h}}) \simeq \check{\mathfrak{h}},$$

with adjoint action and degree two. The map $(u, v) \mapsto u - v$ realizes this isomorphism and carries the image of $\check{\mathfrak{n}}^+ \oplus \check{\mathfrak{n}}^+$ to $\check{\mathfrak{n}}^+$. Therefore

$$\beta(t) = \prod_{\alpha \in R^+(\check{H})} \frac{1 - \alpha(t)}{1 - q^{-1}\alpha(t)}.$$

The signed weighted multiset Θ is $\{(\alpha, +1, 1) : \alpha \in R(\check{H})\}$. Its color cone is the cone generated by the simple roots of $\check{H}$; thus Θ^+ consists exactly of its positive roots. Every root is positive or negative, and a simple reflection changes the sign of exactly its own simple root. This proves both assertions of Statement 7.1.5, with multiplicity one and the displayed grading. A simple spherical root of the group case has support the corresponding simple root in each factor, hence support type $A_1 \times A_1$. Statement 6.3.1 therefore follows from Sakellaridis’s Theorem 6.3.2, including when the whole group H is exceptional. Alternatively the same rank-one coefficient is the usual SL_2 group spherical-function calculation. For its simple reflection,

$$B_{s_\alpha}(t) = -\alpha(t) \frac{1 - q^{-1}\alpha(t)^{-1}}{1 - q^{-1}\alpha(t)} = \frac{\beta(t)}{\beta(s_\alpha t)}.$$

The cocycle relation proves the formula for all w .

Finally, for an irreducible admissible tensor product $\pi_1 \boxtimes \pi_2$, diagonal invariants identify with $\mathrm{Hom}_H(\pi_1, \pi_2^\vee)$, which has dimension at most one by Schur’s lemma. This verifies the multiplicity-one hypothesis for this literal diagonal pair. $\square$

6 Coverage ledger for simple classical ambient groups

The characteristic-zero column of Knop–Röhrle, Table 1 ([arXiv:1305.3183v3](#), p. 3), reproduces Krämer’s connected reductive classification. It contains nineteen displayed rows. Two rows are triality variants of orthogonal block subgroups, leaving seventeen families up to outer automorphism. The trivial subgroup pair $H = G$ is added separately. The following is a coverage ledger; a “retained” entry means that its geometric type is not excluded here by type N, and does not assert local multiplicity one or a completed Euler-factor calculation.

	Connected pair $H \subset G$	BP reference	Scope or qualification
C1	$\mathrm{SO}_n \subset \mathrm{SL}_n, n \geq 2$	5	Type N for $n \geq 3$; $n = 2$ is the torus case.
C2	$\mathrm{S}(\mathrm{GL}_m \times \mathrm{GL}_n) \subset \mathrm{SL}_{m+n}, m \geq n \geq 1$	3	Retained. BP4 is its normalizer when $m = n$, not the connected pair.
C3	$\mathrm{SL}_m \times \mathrm{SL}_n \subset \mathrm{SL}_{m+n}, m > n \geq 1$	augmentation of 3	Retained; central character lattice must be restored.
C4	$\mathrm{Sp}_{2n} \subset \mathrm{SL}_{2n}, n \geq 2$	6	Retained; BP adjoins the ambient center.
C5	$\mathbf{G}_m \cdot \mathrm{Sp}_{2n} \subset \mathrm{SL}_{2n+1}, n \geq 1$	31	Retained; $n = 1$ overlaps C2.
C6	$\mathrm{Sp}_{2n} \subset \mathrm{SL}_{2n+1}, n \geq 1$	augmentation of 31	Retained; $n = 1$ overlaps C3.
C7	$\mathrm{GL}_n \subset \mathrm{Sp}_{2n}, n \geq 1$	12	Type N for $n \geq 2$; $n = 1$ is a torus case.
C8	$\mathbf{G}_m \times \mathrm{Sp}_{2n-2} \subset \mathrm{Sp}_{2n}, n \geq 2$	36	Retained. BP37 is a different, disconnected normalizer.
C9	$\mathrm{Sp}_{2a} \times \mathrm{Sp}_{2b} \subset \mathrm{Sp}_{2a+2b}, a, b \geq 1$	10	Retained. BP11 is a different normalizer when $a = b$.
C10	$\mathrm{GL}_n \subset \mathrm{SO}_{2n}, n \geq 2$	16	Retained; low ranks overlap other rows. BP17 is a normalizer.
C11	$\mathrm{SL}_n \subset \mathrm{SO}_{2n}, \text{odd } n \geq 3$	augmentation of 16	Retained; do not omit this central augmentation.
C12	$G_2 \subset \mathrm{SO}_8$	38	Retained; BP adjoins Z_{SO_8} .
C13	$\mathrm{SO}_2 \times \mathrm{Spin}_7 \subset \mathrm{SO}_{10}$	39	Retained; the SO_2 factor cannot simply be removed.
C14	$\mathrm{GL}_n \subset \mathrm{SO}_{2n+1}, n \geq 2$	32	Retained. BP33 is its disconnected normalizer.
C15	$\mathrm{SO}_m \times \mathrm{SO}_n \subset \mathrm{SO}_{m+n}, m \geq n \geq 1$	7–9, 14–15	Connected subgroup, not $\mathrm{S}(\mathrm{O}_m \times \mathrm{O}_n)$. Subfamilies below.
C16	$\mathrm{Spin}_7 \subset \mathrm{SO}_9$	34	Retained.
C17	$G_2 \subset \mathrm{SO}_7$	35	Retained.

The two extra displayed rows of Knop–Röhrle are $\mathrm{Sp}_4 \otimes \mathrm{Sp}_2 \subset \mathrm{SO}_8$, which is triality-equivalent to $\mathrm{SO}_5 \times \mathrm{SO}_3$, and $\mathrm{Spin}_7 \subset \mathrm{SO}_8$, which is triality-equivalent to $\mathrm{SO}_7 \subset \mathrm{SO}_8$. Their distinct inner-conjugacy classes should be retained if the task is classification up to inner automorphism rather than outer automorphism. Small values making the ambient orthogonal group nonsimple are omitted when this ledger is used specifically for Brion case 1.

The orthogonal block subfamilies. For C15 the sphere cases $n = 1$ are retained. For $n = 2$, the even ambient group is retained, with spherical roots of the schematic forms α_1 and $2\alpha_2 + \cdots + 2\alpha_{r-2} + \alpha_{r-1} + \alpha_r$; the latter is a sum of two orthogonal roots and is type G, despite its coefficients equal to two. In odd ambient dimension, the terminal root is $2(\alpha_2 + \cdots + \alpha_r)$, twice a root of the ambient B_r system, and gives type N. For block sizes $n \geq 3$, a doubled simple root occurs (subject to the usual low-rank identifications), so those subfamilies are excluded. This division is read from the connected spherical systems, with BP’s disconnected normalizers undone; it cannot be read by treating every BP row as connected.

The test for type N is the normalized-root and character-lattice test in Sakellaridis, §6.2: a root-type spherical direction is type T when its normalized ambient root belongs to $\mathcal{X}(X)$, and type N otherwise. Merely searching for a coefficient two, or merely searching for 2α with α simple, is insufficient. The actual character lattice remains part of the datum when passing to central covers or adjoining centers.

Coverage count and missing augmentations. Knop–Röhrle Table 2 has fourteen exceptional characteristic-zero rows, so the two tables give thirty-one nontrivial families up to outer automorphism, before parameter specializations and adding $H = G$. BP’s wonderful list is organized differently. In particular, C3, C6, C11, and the exceptional pair $D_5 \subset E_6$ without its central torus are easily lost if one only copies connected wonderful rows. Conversely, normalizers and finite central enlargements in BP cannot silently replace Krämer’s connected subgroups.

Why the two low-rank torus exceptions are retained. For C1 the eigencharacter lattice is $2P(A_{n-1})$. If $n \geq 3$, the normalized direction of each doubled simple spherical root is an ambient simple root α_i having an odd coefficient on an adjacent fundamental weight. Thus $\alpha_i \notin 2P$, and the root is type N. When $n = 2$ there is only $\alpha_1 = 2\omega_1$; it does lie in the lattice and gives type T. This endpoint is C2 with two one-dimensional blocks. For C7 the lattice is $2P(C_n)$. When $n \geq 2$, the direction α_1 is not in this lattice, because its expression in fundamental weights has an odd adjacent coefficient (for $n = 2$ it is $2\omega_1 - \omega_2$). This gives a type-N root. When $n = 1$, the unique direction is $2\omega_1$ and is type T. The two endpoints are the same split torus pair in $\mathrm{SL}_2 = \mathrm{Sp}_2$; the explicit square-class functionals in Proposition 6 show why they nevertheless fail Assumption 4.5.1 for that simply connected ambient form.

7 The remaining type-A simple families

This section proves the reflection condition for C2, C3, C5 and C6 in the coverage ledger. All multisets retain their multiplicities. The formulas are first for the split homogeneous model; arithmetic descent is discussed separately. Write $R[d]$ for a representation in grading degree d .

7.1 C2 and C3: two general-linear blocks

Put $N = p + q$, $1 \leq p \leq q$, and $\delta = q - p$. For $H = \mathrm{S}(\mathrm{GL}_p \times \mathrm{GL}_q)$ in SL_N the normalized spherical roots are

$$\gamma_i = \alpha_i + \alpha_{N-i} \quad (i < p), \quad \gamma_p = \alpha_p + \cdots + \alpha_q.$$

When $p = q$ the last root is the simple root α_p . The dual Lie algebra is of type C_p . In its standard weight coordinates $e_1, \dots, e_p$, the colors are

$$d_i = e_i - e_{i+1} \quad (i < p), \quad e_p, e_p.$$

The first colors are of type G and have exponent 1; the last two have exponent $(\delta + 1)/2$. These data can also be read from the highest-weight generators $\omega_i + \omega_{N-i}$ ($1 \leq i \leq p$), where the last generator is $2\omega_p$ if $p = q$. The two last colors have the same valuation on the ordinary weight lattice, but must be counted separately. The Levi of $P(X)$ is the middle $A_{\delta-1}$, with the empty-Levi convention for $\delta \leq 1$. Pairing $\rho - \rho_{A_{\delta-1}}$ with the two boundary coroots gives $(\delta + 1)/2$; the type-G colors have pairing 1. This also proves the stated exponents.

The full positive virtual multiset is

$$\Theta^+ = \{(e_i - e_j; 1), (e_i + e_j; 1) : i < j\} \sqcup \{(e_i; (\delta + 1)/2)^{[2]} : 1 \leq i \leq p\}. \quad (\text{A1})$$

Its cone is $C = \mathrm{cone}(d_1, \dots, d_{p-1}, e_p)$. Every positive difference is a sum of adjacent differences, every e_i is $e_p + \sum_{j=i}^{p-1} d_j$, and every positive sum is a sum of two such vectors. The functional with values $p, p-1, \dots, 1$ on the e_i is strictly positive on the cone generators. Thus the full signed-permutation orbits split into Θ^+ and its negative. Interchanging e_i, e_{i+1} crosses exactly d_i ; changing the sign of e_p crosses its two copies and permutes all the other positive weights. This proves Statement 7.1.5.

Here is a direct calculation of the representation factors. Under the Arthur $\mathfrak{sl}_2$ the ambient standard representation is

$$W \oplus V_{\delta-1}, \quad W = \text{Std}_{\text{Sp}_{2p}},$$

where the second summand is absent for $\delta = 0$. The two off-diagonal Hom spaces contribute $W[\delta+1]$ twice when $\delta > 0$. The quotient of $\mathfrak{sl}(W)$ by $\mathfrak{sp}(W)$ is $\bigwedge_0^2 W$ in degree 2. If $\delta = 0$, the two degree-one standard representations instead come from the two even-sphere colors. Consequently, up to zero torus weights,

$$V_X = \bigwedge^2 W[2] \oplus 2W[\delta+1]. \quad (\text{A2})$$

For $\delta = 0$ the second summand belongs to S_X ; otherwise it belongs to the centralizer quotient. The notation $\bigwedge^2 W$ in (A2) records its nonzero weights; its scalar summand is irrelevant here. In the ambient positive-root ordering W has coordinates $e_1, \dots, e_p, -e_p, \dots, -e_1$, with the Arthur block inserted in the middle. The positive nonzero highest-weight lines are exactly (A1).

For C3, $H = \text{SL}_p \times \text{SL}_q$ with $p < q$, retain the additional central coordinate z . Replace the two last colors by $e_p + z, e_p - z$, and replace the standard-weight part of (A1) by

$$\{(e_i + z; (\delta+1)/2), (e_i - z; (\delta+1)/2) : 1 \leq i \leq p\}. \quad (\text{A3})$$

The Weyl group fixes z . All these weights are in the color cone, since $e_i \pm z = \sum_{j=i}^{p-1} d_j + (e_p \pm z)$; also $e_i + e_j$ is a sum of adjacent differences and $(e_p + z) + (e_p - z)$. The preceding strictly positive functional extends by zero on z , proving sign separation. The last reflection crosses exactly $e_p + z, e_p - z$. The two Hom spaces in (A2) carry opposite central characters, which proves the graded comparison as well. Setting $z = 0$ recovers C2. This includes $p = 1$, with the empty difference-root orbit.

7.2 C5 and C6: a symplectic subgroup of odd special-linear degree

Let $G = \text{SL}_{2n+1}$, $n \geq 1$. Start with $H = \text{Sp}_{2n}$, fixing the last one-dimensional complement. The weight monoid is freely generated by $\omega_1, \dots, \omega_{2n}$: in $\bigwedge^i(\mathbf{C}^{2n} \oplus \mathbf{C})$ the invariant is the symplectic exterior power for even i , and that power wedged with the last line for odd i . The corresponding color valuations are the ambient simple coroots. The spherical roots are

$$\gamma_i = \alpha_i + \alpha_{i+1} \quad (1 \leq i < 2n).$$

Write the ambient diagonal coordinates in the order

$$u_1, v_1, u_2, v_2, \dots, u_n, v_n, u_{n+1}.$$

The little Weyl group permutes the u coordinates and the v coordinates separately. Its dual Lie algebra is $\mathfrak{sl}(\mathfrak{gl}_{n+1} \oplus \mathfrak{gl}_n)$. The colors are the $2n$ successive differences in this ordered list; each has exponent 1, since $P(X) = B$.

The virtual multiset consists of every $\pm(u_i - v_j)$, with exponent 1. Its positive half consists of those differences directed from an earlier entry to a later entry of the displayed list. Every such difference is the sum of the intervening adjacent colors. A strictly decreasing functional on the ordered list separates positive from negative weights. Swapping adjacent u_i, u_{i+1} crosses precisely $u_i - v_i$ and $v_i - u_{i+1}$; swapping adjacent v_i, v_{i+1} crosses $v_i - u_{i+1}$ and $u_{i+1} - v_{i+1}$. These pairs are exactly the two virtual colors for the corresponding spherical root. This proves Statement 7.1.5 for C6 uniformly in n .

Here $e = 0$ and $S_X = 0$. The quotient of $\mathfrak{sl}_{2n+1}$ by the displayed block-diagonal dual Lie algebra is

$$(\text{Hom}(\mathbf{C}^n, \mathbf{C}^{n+1}) \oplus \text{Hom}(\mathbf{C}^{n+1}, \mathbf{C}^n))[2].$$

Its positive ambient root spaces give exactly the positive half just listed, proving the required Euler-factor comparison.

For C5 adjoin the torus $\text{diag}(tI_{2n}, t^{-2n})$ to H . The colors descend along the quotient by the Weyl-fixed direction

$$q = \sum_{j=1}^n (-(n-j+1)c_{2j-1} + jc_{2j}), \quad (\text{A4})$$

where c_i are the successive color differences. Indeed the weights ω_{2j} and ω_{2j-1} have torus charges $2j$ and $-2(n-j+1)$; (A4) is the primitive charge vector in the dual space. In ambient diagonal coordinates it is constant $-n$ on the u block and $n+1$ on the v block. Thus the quotient is compatible with the little Weyl group and removes its central direction. The functional taking value 1 on every c_i vanishes on (A4), because the positive and negative coefficients both sum to $n(n+1)/2$. It therefore descends to a strictly positive functional on the new color cone. Every positive-orbit expression above descends, and no negative weight can become positive. The crossing calculation, including all multiplicity labels, is consequently unchanged. The same block Hom decomposition gives the quotient representation after restricting the central character. The possible additional trivial Lie-algebra quotient has no effect on the cocycle. This proves C5, including $n=1$.

Multiplicity-one scope and the low-rank torus entries. The Lie-theoretic verification above does not assert that every listed central form satisfies Assumption 4.5.1. Two concrete exclusions are useful. For C2 with $p=q$, the weight lattice is generated by $\omega_i + \omega_{2p-i}$ for $i < p$ and $2\omega_p$. Its saturation adds ω_p , giving a quotient of order two. Thus the diagonalizable image of the generic Borel stabilizer has a split μ_2 component, whose $H^1(k, \mu_2)$ is nontrivial. By the generic multiplicity formula of Sakellaridis (2008), Theorem 1.2.1, the literal split simply connected pair fails Assumption 4.5.1. At $p=q=1$ this is exactly $\text{SO}_2 \backslash \text{SL}_2$, the low-rank endpoint of Kramer C1; it is also $\text{GL}_1 \backslash \text{Sp}_2$, the endpoint of C7. Both have type T, not type N, and are already included here.

C6 has a different obstruction. Its lattice is the entire ambient weight lattice, and $P(X) = B$, so its admissible torus is the entire dual torus. The normalizer in the ambient Weyl group is consequently all of S_{2n+1} , whereas $W_X = S_n \times S_{n+1}$. The same generic multiplicity formula therefore gives

$$m_{\text{gen}} \geq \frac{(2n+1)!}{n!(n+1)!} > 1.$$

Thus C6 also fails Assumption 4.5.1 for every $n \geq 1$, despite the saturation of its eigencharacter lattice. This demonstrates why checking only lattice saturation is insufficient. Its $n=1$ endpoint is also C3 with $p=1, q=2$, and has generic multiplicity at least three.

8 The family $\text{Sp}_{2n} \backslash \text{SL}_{2n}$

Let $n \geq 2$, let $G = \text{SL}_{2n}$, and let $H = \text{Sp}_{2n}$ in its standard embedding. Number the ambient simple roots $\alpha_1, \dots, \alpha_{2n-1}$ in their chain order. The spherical system (BP case 6, with the connected subgroup and its actual lattice retained) has

$$S^p = \{\alpha_1, \alpha_3, \dots, \alpha_{2n-1}\}, \quad \gamma_i = \alpha_{2i-1} + 2\alpha_{2i} + \alpha_{2i+1} \quad (1 \leq i < n).$$

Its eigencharacter lattice has basis $\lambda_i = \omega_{2i}$, $1 \leq i < n$, and

$$\gamma_i = 2\lambda_i - \lambda_{i-1} - \lambda_{i+1}, \quad \lambda_0 = \lambda_n = 0.$$

Each even simple root moves one color D_i , whose valuation satisfies $\langle v_{D_i}, \lambda_j \rangle = \delta_{ij}$. Consequently the v_{D_i} are the simple roots β_i of a dual root system A_{n-1} . All spherical roots are type G: each γ_i is a sum of two orthogonal ambient roots, and has support A_3 . There are no even-sphere colors, so $S_X = 0$.

The centralizer quotient and its grading. Write the standard $2n$ -dimensional dual vector space as $W \otimes U$, where $\dim W = n$ and U is the standard two-dimensional $\mathfrak{sl}_2$ -module. The Arthur $\mathfrak{sl}_2$ acts on U , the dual Lie algebra $\check{\mathfrak{g}}_X = \mathfrak{sl}(W)$ acts on W , and $e = 1 \otimes E_{12}$. Since $\text{End}(U) = \mathbf{C} 1 \oplus \mathfrak{sl}(U)$, the nilpotent centralizer is

$$\mathfrak{sl}(W \otimes U)^e = (\mathfrak{sl}(W) \otimes 1) \oplus (\text{End}(W) \otimes E_{12}).$$

Indeed the centralizer of E_{12} in $\text{End}(U)$ is spanned by $1, E_{12}$, and the trace-zero condition is exactly $\text{tr}(A) = 0$ on the coefficient of 1 . The first summand is $\check{\mathfrak{g}}_X$, so

$$V_X = \check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X \simeq \text{End}(W).$$

On the remaining highest-weight vectors, $\text{ad}(h)$ has eigenvalue two. The grading prescribed by Definition 4.5.5 is therefore $d = 2 + 2 = 4$. This proves the grading, rather than inferring it from an expected L-factor.

The positive part, including zero weights. Order the basis of $W \otimes U$ blockwise as $(w_1 u_1, w_1 u_2, w_2 u_1, w_2 u_2, \dots)$, so that e is in the positive nilpotent subalgebra. A vector of the centralizer is $A \otimes 1 + B \otimes E_{12}$. It is strictly upper triangular if and only if A is strictly upper triangular and B is upper triangular. Passing to the quotient kills A , and gives

$$V_X^+ = \{B \in \text{End}(W) : B \text{ upper triangular}\}, \quad \text{in degree four.}$$

Thus the nonzero positive weights are $\varepsilon_i - \varepsilon_j$, $i < j$, each once. There are also n zero weights. Their Euler factors are constant in the Satake parameter and cancel in the cocycle. This explicitly verifies the positive quotient construction used in the definition.

Proof of Statement 7.1.5. The half sum $\rho_{P(X)}$ of roots in the unipotent radical of $P(X)$ has blockwise coefficients $(n-1, n-1, n-3, n-3, \dots, 1-n, 1-n)$. Thus $\langle \alpha_{2i}^\vee, \rho_{P(X)} \rangle = 2$, and the virtual weighted colors are $(\beta_i, +1, 2)$, $1 \leq i < n$. Their Weyl closure is

$$\Theta = \{(\varepsilon_i - \varepsilon_j, +1, 2) : i \neq j\}.$$

The color cone is generated by $\beta_i = \varepsilon_i - \varepsilon_{i+1}$. Every positive weight has the exact certificate

$$\varepsilon_i - \varepsilon_j = \beta_i + \dots + \beta_{j-1} \quad (i < j).$$

The cone is pointed, as seen from any functional with $\ell(\varepsilon_1) > \dots > \ell(\varepsilon_n)$. The simple reflection exchanging $\varepsilon_i, \varepsilon_{i+1}$ changes the sign of exactly β_i . It preserves the sign of every other positive root, since it cannot reverse any other pair of distinct indices. The unique crossing is therefore the unique virtual weighted color belonging to γ_i , including its exponent two. This proves both parts of Statement 7.1.5 for every $n \geq 2$.

The theorem's cocycle for this family. Put $t_i = \varepsilon_i(t)$; only the ratios t_i/t_j occur, so the formula is independent of a scalar lift of the dual torus. Directly from the proved quotient and positive part,

$$\beta(t) = (1 - q^{-2})^{-n} \prod_{i < j} \frac{1 - t_i/t_j}{1 - q^{-2}t_i/t_j}.$$

The support A_3 is classical, so Statement 6.3.1 follows from Sakellaridis's Theorem 6.3.2. Its simple-reflection factor is

$$B_{s_i}(t) = -\frac{t_i}{t_{i+1}} \frac{1 - q^{-2}t_{i+1}/t_i}{1 - q^{-2}t_i/t_{i+1}} = \frac{\beta(t)}{\beta(s_i t)}.$$

The cocycle relation proves the equality for every $w \in S_n$. The constant factor from zero weights cancels identically. Hence the combinatorial assertion and the V_X^+ comparison needed for Theorem 4.5.8 are proved uniformly for this family. This calculation does not replace a separate check of Assumption 4.5.1 for a chosen rational form.

9 Independent audit of two symplectic classical families

9.1 $\mathrm{Sp}_{2p} \times \mathrm{Sp}_{2q} \backslash \mathrm{Sp}_{2(p+q)}$

Assume $1 \leq p \leq q$, put $N = p + q$ and $\delta = q - p$. Let $u_i = e_{2i-1} + e_{2i}$ for $1 \leq i \leq p$, and let E_i denote the basis dual to the u_i . The free monoid of regular highest weights is

$$\lambda_j = \omega_{2j} = u_1 + \cdots + u_j, \quad 1 \leq j \leq p.$$

Thus the color valuations are $d_i = E_i - E_{i+1}$ ($i < p$) and $d_p = E_p$. The normalized spherical roots are $\gamma_i = u_i - u_{i+1}$ ($i < p$) and $\gamma_p = u_p$; the dual group has root system C_p , with simple roots $E_i - E_{i+1}$ and $2E_p$. The Levi of $P(X)$ has the p factors GL_2 on these pairs and one $\mathrm{Sp}_{2\delta}$ factor on the remaining coordinates. Therefore

$$\rho_{P(X)} = \sum_{i=1}^p (N - 2i + \tfrac{3}{2}) u_i.$$

In particular the original color exponents are

$$r(d_i) = 2 \ (i < p), \quad r(d_p) = \delta + \tfrac{3}{2}.$$

The last spherical root has height $2\delta + 2$, so its companion is again E_p , now with exponent $\delta + \tfrac{1}{2}$. The former roots have type G . Consequently the entire weighted Weyl multiset is

$$\{\pm E_i \pm E_j : i < j\}_2 \sqcup \{\pm E_i\}_{\delta+1/2} \sqcup \{\pm E_i\}_{\delta+3/2}. \quad (\text{C1})$$

The cone generated by the colors is the cone generated by $E_1 - E_2, \dots, E_{p-1} - E_p, E_p$. In (C1) the positive weights are $E_i - E_j, E_i + E_j$ ($i < j$) and E_i at both exponents. For example $E_i = (E_i - E_{i+1}) + \cdots + E_p$, and every listed sum is therefore in the cone. Their negatives cannot lie in it, since that cone is pointed. This accounts for every weight. A simple transposition $E_i \leftrightarrow E_{i+1}$ crosses precisely $E_i - E_{i+1}$. The last simple reflection, $E_p \mapsto -E_p$, exchanges the positive weights $E_i - E_p$ and $E_i + E_p$ and crosses precisely the two occurrences of E_p . Thus Statement 7.1.5 holds for every p, q .

This calculation fixes the shifts in the representation formula. Write $W = \mathrm{Std}_{\mathrm{Sp}_{2p}}$ and let A denote the Arthur SL_2 . On the standard representation of the ambient dual SO_{2N+1} the distinguished map gives

$$(W \otimes \mathrm{Sym}^1 \mathbf{2}_A) \oplus \mathrm{Sym}^{2\delta} \mathbf{2}_A.$$

Taking its exterior square gives an adjoint $\mathfrak{sp}_{2p}$ term at Arthur weight zero, a $\bigwedge^2 W$ term at highest Arthur weight two, and mixed terms with highest Arthur weights $2\delta + 1$ and $2\delta - 1$ (the latter only if $\delta \geq 1$). Hence

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (\bigwedge^2 W)[4] \oplus W[2\delta + 3] \oplus \begin{cases} W[2\delta + 1], & \delta \geq 1, \\ 0, & \delta = 0, \end{cases} \oplus \bigoplus_{j=1}^{\delta} \mathbf{1}[4j]. \quad (\text{C2})$$

When $\delta = 0$, the final simple spherical root is of even-sphere type and $S_X = W[1]$. When $\delta \geq 1$ there are no even-sphere colors and $S_X = 0$. Nonzero weights of (C2), together with S_X , give (C1) with $r = d/2$ exactly. The p zero weights in $\bigwedge^2 W$ and the explicitly trivial summands contribute only zeta factors to an ambient-positive choice of V_X^+ ; these cancel in every cocycle ratio.

In particular the short-root orbit has exponent *two*, not one, and the two standard factors occur at $\delta + \tfrac{1}{2}, \delta + \tfrac{3}{2}$, not at $\tfrac{1}{2}, \tfrac{3}{2}$ for unequal p, q . A table entry suggesting otherwise cannot be used for this case. Already $p = 1$ is independently confirmed by Sakellaridis's formula (6.16), where $n = p + q$: its two exponents are $n - \tfrac{3}{2}$ and $n - \tfrac{1}{2}$. For $p \geq 2$, the calculation of $\langle E_i - E_{i+1}, \rho_{P(X)} \rangle = 2$ checks the exterior-square shift directly.

9.2 $\mathbb{G}_m \times \mathrm{Sp}_{2n-2} \backslash \mathrm{Sp}_{2n}$

For $n \geq 2$ the roots are $\alpha = e_1 - e_2$ and $\gamma = e_1 + e_2$. Use the dual fundamental weights x, y , so that the simple dual roots are $2x, 2y$. The extended highest-weight monoid has free generators (ω_1, χ) , $(\omega_1, -\chi)$, and $(\omega_2, 0)$, where χ is a primitive character of the stabilizer torus. The ordinary monoid is generated by $2\omega_1, \omega_2$. The three color valuations, retaining multiplicities, are

$$x, \quad x, \quad y - x.$$

As $\rho_{P(X)} = \omega_1 + (n-1)\omega_2$, their exponents are $1/2, 1/2, n-1$. The companion of $y - x$ is $y + x$ with exponent $n-1$. Thus

$$\Theta = 2\{\pm x\}_{1/2} \sqcup \{\pm y \pm x\}_{n-1}, \quad \Theta^+ = 2\{x\}_{1/2} \sqcup \{y - x, y + x\}_{n-1}. \quad (\text{C3})$$

The two generators x and $y - x$ span the color cone, and $y + x = (y - x) + 2x$. Changing the sign of x crosses just the two occurrences of x ; changing the sign of y crosses just $y - x, y + x$. This proves Statement 7.1.5 with the repeated-color multiplicity retained. Here

$$S_X = (\mathbf{2}_x \oplus \mathbf{2}_x)[1], \quad \check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (\mathbf{2}_x \otimes \mathbf{2}_y)[2n-2] \oplus \bigoplus_{j=1}^{n-2} \mathbf{1}[4j]. \quad (\text{C4})$$

The ordinary weight lattice has index two in its saturation. Thus the same Borel-stabilizer argument used for the product families proves that this exact split simply connected model fails the manuscript's generic local multiplicity-one hypothesis. The all-positive calculation (C3) is a statement for the standard split stabilizer torus; a nonsplit torus requires the appropriate Frobenius signs on the repeated summands. For $n = 1$ the case is the split torus in SL_2 : delete y and the quotient from (C3)–(C4), leaving the two standard summands.

10 Two orthogonal families: cone and grading calculations

The spherical systems in this section are those of Bravi–Pezzini, cases 16 and 32. The argument is on the actual color cone; the exact central form must still be restored from the full weight lattice.

10.1 $\mathrm{GL}_n \backslash \mathrm{SO}_{2n}$ and the odd SL_n variant

Put $p = \lfloor n/2 \rfloor$, let $u_i = e_{2i-1} + e_{2i}$, and let E_i be dual to u_i . For GL_n the highest-weight cone is generated by $u_1, u_1 + u_2, \dots, u_1 + \dots + u_p$. The colors are

$$d_i = E_i - E_{i+1} \quad (i < p), \quad d_+ = d_- = E_p. \quad (\text{D1})$$

For $n = 2p$, the two final colors are moved by the simple spherical root $e_{2p-1} + e_{2p}$. For $n = 2p + 1$ they are moved by the two end roots of D_n and belong to the spherical root $e_{2p-1} + e_{2p}$, whose support is $D_3 = A_3$. The other normalized spherical roots are $u_i - u_{i+1}$. The dual root system is C_p . The parabolic $P(X)$ has p mutually orthogonal A_1 Levi roots $e_{2i-1} - e_{2i}$. Subtracting their half-sum from the ambient D_n half-sum gives

$$\rho_{P(X)} = \sum_{i=1}^p (n - 2i + \tfrac{1}{2}) u_i.$$

Consequently $r(d_i) = 2$, while $r(d_{\pm}) = 1/2$ for even n and $r(d_{\pm}) = 3/2$ for odd n . The virtual Weyl multiset is

$$\Theta = \{\pm E_i \pm E_j : i < j\}_2 \sqcup 2\{\pm E_i\}_r, \quad r = \begin{cases} 1/2 & n \text{ even,} \\ 3/2 & n \text{ odd.} \end{cases} \quad (\text{D2})$$

The positive weights are $E_i \pm E_j$ for $i < j$ and both copies of E_i . The adjacent transposition $E_i \leftrightarrow E_{i+1}$ crosses only $E_i - E_{i+1}$; the end reflection $E_p \mapsto -E_p$ crosses the two copies of E_p . These are exactly (D1), with their respective exponents. The cone is generated by $E_i - E_{i+1}$ and E_p , so it contains all the listed positive weights and none of their negatives. This proves both parts of Statement 7.1.5 for this family.

For completeness the grading can be computed without reading it from an L -function table. Write $W = \text{Std}_{\text{Sp}_{2p}}$. For even n the ambient dual standard representation is $W \otimes \mathbf{2}_A$, where A is the Arthur SL_2 . Thus

$$S_X = (W \oplus W)[1], \quad \check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (\bigwedge^2 W)[4]. \quad (\text{D3})$$

Indeed $\bigwedge^2(W \otimes \mathbf{2}_A)$ decomposes as $\text{Sym}^2 W \otimes \bigwedge^2 \mathbf{2}_A$ and $\bigwedge^2 W \otimes \text{Sym}^2 \mathbf{2}_A$; the former is precisely the dual adjoint module and the latter has highest Arthur weight two. For odd n the ambient standard representation is $(W \otimes \mathbf{2}_A) \oplus \mathbf{1}^2$; hence

$$S_X = 0, \quad \check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (\bigwedge^2 W)[4] \oplus (W \oplus W)[3] \oplus \mathbf{1}[2]. \quad (\text{D4})$$

The last trivial summand is the Cartan of the complementary orthogonal plane and is excluded by an ambient positive-root convention. The zero weights in $\bigwedge^2 W$ only affect a constant zeta factor. These decompositions reproduce (D2) exactly on nonzero weights. In particular the exterior-square factor occurs at $s = 2$. The $s = 1$ printed in row 6 of the arXiv version of Sakellaridis's appendix is inconsistent with this direct color computation and with (D3)–(D4).

If $n = 2p + 1$ and the stabilizer is SL_n , add a central coordinate Z to the dual weight space. The two final colors become $E_p + Z, E_p - Z$; the preceding colors d_i are unchanged. Their exponent is still $3/2$. The additional weighted Weyl orbits are $\{\pm E_i + Z\}_{3/2}$ and $\{\pm E_i - Z\}_{3/2}$, with positive weights $E_i + Z, E_i - Z$. The same adjacent-transposition argument applies, and the end reflection crosses exactly $E_p + Z, E_p - Z$. All positive weights belong to the color cone because $E_i \pm Z = d_i + \dots + d_{p-1} + (E_p \pm Z)$. This proves Statement 7.1.5 in the central direction as well. The dual Lie algebra is $\mathfrak{sp}_{2p} \oplus \mathbb{C}$, $S_X = 0$, and the nonzero quotient weights are

$$(\bigwedge^2 W)[4] \oplus (W \otimes (\chi \oplus \chi^{-1}))[3], \quad (\text{D5})$$

where the weight of χ is Z . The extra Cartan in (D4) is now part of $\check{\mathfrak{g}}_X$. Spin-versus-special-orthogonal conventions can rescale the integral central generator; (D5)'s Z must be interpreted in the actual character lattice, rather than chosen from a Lie-algebra table.

10.2 $\text{GL}_n \backslash \text{SO}_{2n+1}$

Use the usual B_n coordinates $e_1, \dots, e_n$ and their dual coordinates $E_1, \dots, E_n$. Here $P(X) = B$, so the Arthur SL_2 is trivial. The normalized spherical roots are

$$e_i - e_{i+2} \quad (1 \leq i \leq n-2), \quad e_{n-1}, \quad e_n.$$

Thus the dual roots on odd and even indices form two symplectic systems. The colors are

$$d_i = E_i - E_{i+1} \quad (i < n), \quad E_n, E_n,$$

with exponents 1 for d_i and $1/2$ for the two last colors, since $\rho = \sum_{i=1}^n (n-i+1/2)e_i$. The entire weighted Weyl multiset is

$$\Theta = \{\pm E_i \pm E_j : i \not\equiv j \pmod{2}\}_1 \sqcup 2\{\pm E_i : i \equiv n \pmod{2}\}_{1/2}, \quad (\text{D6})$$

where the first set counts each unordered pair once. The color cone is generated by the adjacent differences and E_n . The positive mixed weights are $E_i - E_j, E_i + E_j$ with $i < j$ of opposite parities;

the positive standard weights are the two copies of E_i of parity n . These exhaust exactly one of each opposite pair. The reflection swapping E_i and E_{i+2} crosses precisely

$$E_i - E_{i+1} = d_i, \quad E_{i+1} - E_{i+2} = d_{i+1}.$$

The reflection changing the sign of E_{n-1} crosses $E_{n-1} - E_n, E_{n-1} + E_n$, the latter being the virtual companion with exponent one. Finally $E_n \mapsto -E_n$ crosses just the two occurrences of E_n , and exchanges all remaining positive mixed weights in pairs. This is the complete check of Statement 7.1.5.

Let W_o, W_e denote the standard modules on odd and even indices. Then

$$\check{\mathfrak{g}}_X = \mathfrak{sp}(W_o) \oplus \mathfrak{sp}(W_e), \quad S_X = (W_{\text{parity } n} \oplus W_{\text{parity } n})[1], \quad \check{\mathfrak{g}}/\check{\mathfrak{g}}_X = (W_o \otimes W_e)[2].$$

This follows at once by taking $\text{Sym}^2(W_o \oplus W_e)$, the adjoint representation of the ambient dual symplectic group. It gives the exponents in (D6) exactly. Unlike the preceding two families, the exterior-square shift issue does not arise here: the mixed term has Arthur highest weight zero and degree two.

11 The orthogonal block families that have no roots of type N

Among the connected pairs $\text{SO}_a \times \text{SO}_b \subset \text{SO}_{a+b}$, with $a \geq b$, the remaining families are the spheres $b = 1$ and, when $a + b = 2n$, the rank-two family $b = 2$. The following calculations use the connected subgroup. Replacing it by its normalizer changes the weight lattice and can change the type of a spherical root. Small reducible ambient groups are interpreted through their usual product identifications. We write $\doteq$ for equality up to a nonzero factor independent of the dual-torus parameter. Such factors cancel from every cocycle. In particular, this notation removes the harmless ambiguity in which zero-weight vectors are included in a positive part.

11.1 Even-dimensional spheres

Let $X = \text{SO}_{2n} \backslash \text{SO}_{2n+1}$, $n \geq 2$. Its spherical root is $\gamma = \varepsilon_1$, its dual root is $2u$, and its single color has valuation u . The Levi of $P(X)$ has type B_{n-1} , and

$$\rho_{P(X)} = (n - \tfrac{1}{2})\varepsilon_1.$$

The relevant color therefore has exponent $n - \frac{1}{2}$. Furthermore $\langle \check{\rho}, \gamma \rangle = n$, so the complementary virtual color in Sakellaridis's recipe has exponent $\frac{1}{2}$. Thus, for the split stabilizer, the complete nonzero weighted multiset is

$$\Theta = \{(\pm u, +, \tfrac{1}{2}), (\pm u, +, n - \tfrac{1}{2})\}. \quad (\text{O1})$$

Here and below a displayed $\pm$ denotes both signs, each once. The color cone is $\mathbf{R}_{\geq 0}u$. Its positivity and the reflection-crossing condition are immediate: the nonidentity Weyl element sends both positive occurrences of u to $-u$, and these are precisely the two virtual colors. This proves Statement 7.1.5 for the entire family.

For completeness the representation comparison follows directly from matrices. On the standard symplectic space for $\check{G} = \text{Sp}_{2n}$, the commuting pair $\check{G}_X \times \text{SL}_{2, \text{Arthur}}$ acts on

$$U \oplus W, \quad U = \text{Std}_{\text{SL}_2}, \quad W = \text{Sym}^{2n-3} \mathbf{C}^2.$$

The first group acts on U and the Arthur group on W . Using $\mathfrak{sp}(U \oplus W) = \text{Sym}^2(U \oplus W)$,

$$\check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X = U[2n-1] \oplus \bigoplus_{j=1}^{n-1} \mathbf{C}[4j]. \quad (\text{O2})$$

Indeed, the highest Arthur weight in W is $2n - 3$, while the principal centralizer in $\mathfrak{sp}(W)$ has highest weights $2, 6, \dots, 4n - 6$; the definition of the grading adds two. The even-sphere color gives $S_X = U[1]$. Thus its nonzero positive weights are exactly (O1). The positive vector in each copy of U is the u -weight vector: in an ambient orthogonal/symplectic root basis, the first coordinate selects this sign. If $z = u(t)$,

$$\beta(t) \doteq \frac{1 - z^2}{(1 - q^{-1/2}z)(1 - q^{-(n-1/2)}z)}. \quad (\text{O3})$$

All spherical-root support is classical, so Sakellaridis's rank-one calculation applies and the crossing lemma proves the required cocycle. The endpoint $n = 1$ is the torus quotient of SO_3 ; there are two colors, $S_X = 2U[1]$, and the Lie quotient is zero. Formula (O3) still holds at this endpoint, although the derivation of (O2) does not apply.

There is a printed discrepancy in the cited source that should not be propagated: Sakellaridis (2013), equation (6.13), prints q^{-n} in place of $q^{-(n-1/2)}$. The exponent $n - 1/2$ follows from his own virtual-color definition in §7.1, from (O2), from the pairing $2n - 1$ computed in BZSV §4.7, and from the corrected rank-one table in Sakellaridis, *Functorial transfer between relative trace formulas in rank one*, arXiv:1808.09358v2 (2023), p. 4.

The nonsplit stabilizer inside split G . For an unramified nonsplit even orthogonal stabilizer, the single actual color is still defined over the field, and its exponent $n - 1/2$ has Frobenius sign $+$. The formal even-sphere color defining S_X is the difference of the two terminal geometric Borel orbits; Frobenius exchanges those orbits. Consequently it acts by -1 on that line and, by compatibility with the split dual SL_2 , on the whole copy $U[1]$. This is exactly the simple Galois action of BZSV Definition 4.8.9. The Lie quotient (O2) has trivial Frobenius, since its ambient split pinning and Arthur map are defined over the base field. Thus replace the first factor of the denominator in (O3) by $1 + q^{-1/2}z$. Equivalently, (O1) has sign $-$ in degree one and sign $+$ in degree $2n - 1$. The signed crossing identities remain valid. At $n = 1$, the two degree-one copies instead have Frobenius eigenvalues $+1, -1$, as required for a nonsplit torus.

11.2 Odd-dimensional spheres

Let $X = \text{SO}_{2n-1} \backslash \text{SO}_{2n}$, $n \geq 2$. The normalized spherical root is $2\varepsilon_1$, the dual root and color valuation are both u , and the color exponent is $n - 1$ because $\rho_{P(X)} = (n - 1)\varepsilon_1$. This is a root of type G, not type N: its support expresses it as the sum of the two orthogonal roots $\varepsilon_1 - \varepsilon_n$ and $\varepsilon_1 + \varepsilon_n$. There are no even-sphere colors, so $S_X = 0$, and

$$\Theta = \{(u, +, n - 1), (-u, +, n - 1)\}, \quad C = \mathbf{R}_{\geq 0}u. \quad (\text{O4})$$

Again the sole reflection crosses exactly the one positive virtual color.

The standard orthogonal representation of $\check{G} = \text{SO}_{2n}$ restricts to the commuting pair as

$$U \oplus W, \quad U = \text{Std}_{\text{SO}_3}, \quad W = \text{Sym}^{2n-4} \mathbf{C}^2.$$

The trivial line in the standard representation of the principal SL_2 in SO_{2n-2} , together with the two coordinates of the split dual torus, gives U . Therefore

$$\mathfrak{so}(U \oplus W) = \bigwedge^2 U \oplus (U \otimes W) \oplus \bigwedge^2 W,$$

and removing $\check{\mathfrak{g}}_X = \bigwedge^2 U$ gives

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = U[2n - 2] \oplus \bigoplus_{j=1}^{n-2} \mathbf{C}[4j]. \quad (\text{O5})$$

The nonzero weights of U are $\pm u$, and the positive ambient root spaces select $+u$. Hence, for $z = u(t)$,

$$\beta(t) \doteq \frac{1-z}{1-q^{-(n-1)}z}. \quad (\text{O6})$$

The formula includes $n = 2$, when W is one-dimensional, the Arthur group is trivial, and the quotient is the adjoint representation of SO_3 in degree two. The classical rank-one calculation is Sakellaridis (2013), equation (6.18). For split even-dimensional ambient quadratic space the stabilizer of a nonsingular vector is a split odd orthogonal group: a maximal isotropic subspace intersected with the perpendicular hyperplane has dimension at least $n - 1$. Thus this family has no additional nonsplit-stabilizer case.

11.3 The rank-two even orthogonal blocks

Let

$$X = (\text{SO}_2 \times \text{SO}_{2n-2}) \backslash \text{SO}_{2n}, \quad n \geq 3.$$

The simple spherical roots are

$$\gamma_1 = \varepsilon_1 - \varepsilon_2 = \alpha_1, \quad \gamma_2 = 2\varepsilon_2 = 2\alpha_2 + \cdots + 2\alpha_{n-2} + \alpha_{n-1} + \alpha_n.$$

The first is type T and the second is type G. Their dual roots form B_2 , with simple roots $a = u_1 - u_2$, $b = u_2$. There are two colors with valuation $h = (u_1 - u_2)/2$, and one with valuation u_2 . Since

$$\rho_{P(X)} = (n-1)\varepsilon_1 + (n-2)\varepsilon_2,$$

the first two have exponent $1/2$, and the last has exponent $n - 2$. The full virtual multiset is therefore

$$\Theta = 2\{(\frac{\pm u_1 \pm u_2}{2}, +, \frac{1}{2})\} \sqcup \{(\pm u_i, +, n-2) : i = 1, 2\}. \quad (\text{O7})$$

The actual dual group is the central form that admits the spin weights; its Lie algebra is $\mathfrak{so}_5$. Coordinates on a covering torus are harmless, provided the weights are descended on the actual character lattice when applying the formula.

Here is a complete cone and crossing certificate. The color cone is $C = \mathbf{R}_{\geq 0}h + \mathbf{R}_{\geq 0}u_2$, and its positive weights, with their nonnegative expressions, are

$$h, \quad (u_1 + u_2)/2 = h + u_2, \quad u_2, \quad u_1 = 2h + u_2. \quad (\text{O8})$$

Every weight in (O7) is one of these or its negative. The functional $\ell(u_1) = 2, \ell(u_2) = 1$ is strictly positive on both generators. The reflection s_a exchanges u_1, u_2 ; it flips only h , with multiplicity two and exponent $1/2$. The reflection s_b negates u_2 ; it interchanges the two positive spin weights and flips only u_2 , with exponent $n - 2$. These are exactly the virtual colors for γ_1 and γ_2 . This proves Statement 7.1.5 without a rank bound.

To compare the representation, the two type-T colors give $S_X = 2\text{Spin}_5[1]$, where Spin_5 here means the four-dimensional spin representation. The standard $2n$ -dimensional representation of $\check{G}$ decomposes as

$$U \oplus W, \quad U = \text{Std}_{\text{SO}_5}, \quad W = \text{Sym}^{2n-6} \mathbf{C}^2.$$

Indeed the principal Arthur group in the Levi SO_{2n-4} acts on its standard representation as $W \oplus \mathbf{C}$, and the remaining four torus coordinates with this fixed line form U . The same exterior-square calculation as above yields

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = \text{Std}_{\text{SO}_5}[2n-4] \oplus \bigoplus_{j=1}^{n-3} \mathbf{C}[4j]. \quad (\text{O9})$$

Its nonzero weights are $\pm u_1, \pm u_2$; in the compatible ambient Borel its positive part contains u_1, u_2 . The spin positivity is (O8), so (O7) is precisely the multiset of nonzero graded weights of V_X , with the asserted positive part. Consequently

$$\beta(t) \doteq \frac{\prod_{v \in \{u_1 - u_2, u_1 + u_2, u_1, u_2\}} (1 - v(t))}{(1 - q^{-1/2}h(t))^2 (1 - q^{-1/2}(h + u_2)(t))^2 (1 - q^{-(n-2)}u_1(t))(1 - q^{-(n-2)}u_2(t))}. \quad (\text{O10})$$

Both supports are classical: A_1 for γ_1 and D_{n-1} for γ_2 , with $D_2 = A_1 \times A_1$ at the endpoint $n = 3$. Thus the rank-one analytic hypotheses and the representation comparison are established in every rank. The crossing lemma proves the full Weyl cocycle.

If the orthogonal two-plane is unramified anisotropic, Frobenius exchanges its two isotropic lines and therefore the two type-T colors. On the multiplicity space of the two spin representations the eigenvalues are $+1, -1$; the Lie quotient (O9) retains trivial Frobenius for the split ambient pinning. Replace each squared spin factor in (O10) by $(1 - q^{-1/2}v(t))(1 + q^{-1/2}v(t))$, with $v = h, h + u_2$. The preceding cone proof and signed crossing proof are unchanged. This verifies the simple unramified Galois action for these rational forms as well.

These are geometric and signed Euler-factor proofs. In applying the manuscript's theorem one must additionally retain Assumption 4.5.1 for the specified central form and field; none of the cone calculations is a proof of multiplicity one for every irreducible unramified representation.

A concrete multiplicity obstruction for the rank-two blocks. For the literal SO_{2n} form, the eigencharacter lattice is

$$\Lambda_X = \mathbf{Z}(2\varepsilon_1) + \mathbf{Z}(\varepsilon_1 + \varepsilon_2) = \{a\varepsilon_1 + b\varepsilon_2 : a + b \text{ is even}\}. \quad (\text{O11})$$

Here is a direct verification, avoiding a normalizer table. The rank-two spherical roots place the rational span in $\mathbf{Q}\varepsilon_1 + \mathbf{Q}\varepsilon_2$. The subgroup contains the central element $-I$ of SO_{2n} , so any irreducible representation with an invariant vector has even coordinate sum. Conversely, the traceless quadratic tensor measuring the two orthogonal summands supplies an invariant in highest weight $2\varepsilon_1$, and the volume bivector of the two-plane supplies one in highest weight $\varepsilon_1 + \varepsilon_2$. Sphericity then identifies these highest weights with eigencharacters, and they generate the entire even-sum lattice in that span. This proves (O11).

The saturation in the ambient torus character lattice is $\mathbf{Z}\varepsilon_1 + \mathbf{Z}\varepsilon_2$, so the quotient has order two. Equivalently the diagonalizable image of a generic Borel stabilizer has a μ_2 factor. For a split ambient group the number of rational orbits in the geometric open Borel orbit therefore has a factor $H^1(k, \mu_2) = k^\times / (k^\times)^2$, which is nontrivial over every nonarchimedean local field of characteristic zero. Sakellaridis, arXiv:math/0606130, Theorem 1.2.1, consequently implies that generic unramified multiplicity exceeds one. Thus the literal SO_{2n} pair, and also its simply connected preimage, are outside Assumption 4.5.1. For the preimage the saturation remains the integer lattice in this two-coordinate span, since a half-integral weight of type D_n has nonzero half-integral coordinates in every position and $n \geq 3$. Passing to the adjoint ambient form changes this argument: its character lattice is the even-sum root lattice, in which (O11) is saturated. No failure of multiplicity one for that adjoint form follows from (O11).

12 Exceptional stabilizers in classical ambient groups

This section supplies four individual checks which are easy to miss when the classical families are organized only by matrix-block embeddings. The ambient groups are taken simply connected as in Theorem 1.1.28. Orthogonal notation below specifies the Lie-theoretic embedding; it does not authorize replacing a connected lifted stabilizer by its full, possibly disconnected, inverse image. All spherical-root supports here are classical, so [Sak13, Theorem 6.3.2] applies once the other local hypotheses of that paper hold.

$G_2 \subset \text{Spin}_7$. The spherical root is $\gamma = \alpha_1 + 2\alpha_2 + 3\alpha_3$, with $S^p = \{\alpha_1, \alpha_2\}$; see [BP15, case 35]. It is of type G. There is one color, and $\Theta^+ = \{(\check{\gamma}; 3)\}$. For example, $\rho_{A_2} = \alpha_1 + \alpha_2$ and $\langle \alpha_3^\vee, \rho - \rho_{A_2} \rangle = 3$. The rank-one sign argument proves Statement 7.1.5. There are no even-sphere colors, so $S_X = 0$. Write V_j for the Arthur- $\mathfrak{sl}_2$ module of highest weight j , and let A be a two-dimensional multiplicity space. The standard module of $\check{G} = \text{Sp}_6$ is $V_2 \otimes A$. Hence

$$\mathfrak{sp}_6 = (V_4 \oplus V_0) \otimes \text{Sym}^2 A \oplus V_2 \otimes \mathbf{1}.$$

The V_0 term is $\check{\mathfrak{g}}_X$ and consequently

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = \text{Ad}_{A_1}[6] \oplus \mathbf{1}[4].$$

Its unique nonzero positive weight contributes exponent 3, as required. The connected cover Spin_7 / G_2 has the larger character lattice generated by ω_3 ; the normalized root is $2\omega_3$. The adjoint representation descends through the resulting dual isogeny.

$\text{Spin}_7 \subset \text{Spin}_9$. Here the embedding is the spin embedding through Spin_8 , not the ordinary vector-block embedding. In type B_4 , put

$$\gamma = \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4, \quad \delta = \alpha_2 + 2\alpha_3 + 3\alpha_4.$$

The subsystem $S^p = \{\alpha_2, \alpha_3\}$ has type A_2 and the two spherical roots are orthogonal; see [BP15, case 34]. Write t, u for the fundamental weights of the two dual rank-one factors. The two actual colors, moved by α_1 and α_4 , have valuations $t - u$ and $2u$, with exponents 2 and 3. The virtual companion for γ is $t + u$ at exponent 2, since $\text{ht}(\gamma) = 4$. Thus

$$C = \mathbb{R}_{\geq 0}(t - u) + \mathbb{R}_{\geq 0}(2u), \quad \Theta^+ = \{(t - u; 2), (t + u; 2), (2u; 3)\}.$$

The Weyl group changes the signs of t, u independently. Its first simple reflection inverts $t - u, t + u$, and its second inverts $2u$ only. The full multiset is precisely the displayed set and its negative. This proves Statement 7.1.5.

There is an important distinction here: the root γ has full B_4 support, and its localization has rank 2. It is therefore not a rank-one even-sphere localization in the sense used to define S_X . In fact $S_X = 0$. For the dual group Sp_8 , its standard module is $(V_2 \otimes A) \oplus B$, where A, B are two-dimensional symplectic spaces. Taking its symmetric square gives

$$\check{\mathfrak{g}}_X = \mathfrak{sp}(A) \oplus \mathfrak{sp}(B), \quad \check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = \text{Ad}_{\mathfrak{sp}(A)}[6] \oplus (A \otimes B)[4] \oplus \mathbf{1}[4].$$

The three positive nonzero weights have precisely the exponents 3, 2, 2. For the simply connected ambient group the connected stabilizer gives an augmentation of the SO_9 character lattice. The tensor $A \otimes B$ and the adjoint representation descend under the actual dual central quotient. The individual weights t, u need not be characters; only the displayed tensor and adjoint weights are used.

$G_2 \subset \text{Spin}_8$. Use type D_4 numbering with central node 2 and leaves 1, 3, 4. The roots are

$$\gamma_1 = \alpha_1 + \alpha_2 + \alpha_3, \quad \gamma_2 = \alpha_1 + \alpha_2 + \alpha_4, \quad \gamma_3 = \alpha_2 + \alpha_3 + \alpha_4,$$

with $S^p = \{\alpha_2\}$; see [BP15, case 38] and its connected augmentation. They are mutually orthogonal, so the little Weyl group is $(\mathbb{Z}/2)^3$, not the Weyl group of a simple A_3 system. In fundamental-weight coordinates e_1, e_2, e_3 for the three dual rank-one factors, the three color valuations are

$$c_1 = e_1 + e_2 - e_3, \quad c_2 = e_1 - e_2 + e_3, \quad c_3 = -e_1 + e_2 + e_3.$$

Their exponents are all $3/2$, from $\rho_{P(X)} = \rho - \alpha_2/2$. The full orbit consists of the eight weights $\pm e_1 \pm e_2 \pm e_3$, and

$$\Theta^+ = \{(c_1; 3/2), (c_2; 3/2), (c_3; 3/2), (e_1 + e_2 + e_3; 3/2)\}.$$

The fourth weight equals $c_1 + c_2 + c_3$; the functional which sums the three coordinates is positive on this cone and excludes their negatives. Changing the sign of coordinate i inverts exactly the two color weights whose i -coordinate is positive. This proves Statement 7.1.5.

Here $S_X = 0$. The standard orthogonal module of the dual D_4 group is $(V_1 \otimes A) \oplus (B \otimes C)$ with $\dim A = \dim B = \dim C = 2$. Taking its exterior square gives

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (A \otimes B \otimes C)[3] \oplus \mathbf{1}[4].$$

This proves the four required factors at exponent $3/2$. For $G = \text{Spin}_8$, the actual dual group can be described as

$$(\text{SL}_2^3) / \{(\epsilon_1, \epsilon_2, \epsilon_3) : \epsilon_i = \pm 1, \epsilon_1 \epsilon_2 \epsilon_3 = 1\}.$$

The triple tensor representation descends to this quotient. This explicitly retains the connected G_2 rather than replacing it by $G_2 Z(G)$.

$\text{SO}_2 \cdot \text{Spin}_7 \subset \text{Spin}_{10}$. In the type D_5 diagram the trivalent node is 3. There are four spherical roots, not three:

$$\beta_1 = \alpha_2 + \alpha_3 + \alpha_4, \quad \beta_2 = \alpha_1, \quad \beta_3 = \alpha_2 + \alpha_3 + \alpha_5, \quad \eta = \alpha_3 + \alpha_4 + \alpha_5.$$

The first three form an A_3 chain and η is orthogonal to it; $S^p = \{\alpha_3\}$. These data follow from [BP15, case 39]. Let f_1, f_2, f_3, f_4 , with sum 0, be the weights of the standard SL_4 module, so the simple roots are $f_i - f_{i+1}$. Let u be the fundamental weight of the remaining SL_2 . The five color valuations are

$$A = f_2, \quad B = -f_3, \quad C = f_1 + f_3 - u, \quad D = f_1 + f_4 + u, \quad E = f_2 + f_3 + u.$$

The first two are the type-T colors moved by α_1 , of exponent $1/2$; the remaining three, moved by $\alpha_2, \alpha_4, \alpha_5$, have exponent $3/2$. The exponents follow directly from $\rho_{P(X)} = \rho - \alpha_3/2$. The full weighted Weyl orbit is

$$\{(\pm f_i; 1/2) : 1 \leq i \leq 4\} \sqcup \{(f_i + f_j \pm u; 3/2) : i < j\}.$$

Its positive part is exactly

$$\begin{aligned} \Theta^+ = & \{(f_1; 1/2), (f_2; 1/2), (-f_3; 1/2), (-f_4; 1/2)\} \\ & \sqcup \{(f_1 + f_2 \pm u; 3/2), (f_1 + f_3 \pm u; 3/2), (D; 3/2), (E; 3/2)\}. \end{aligned}$$

Here is a proof of the cone assertion without a numerical cone solver. Since $D + E = 2u$, the four new expressions are generated positively using

$$f_1 = C + B + (D + E)/2, \quad -f_4 = C + A + (D + E)/2,$$

$$f_1 + f_2 - u = C + A + B, \quad f_1 + f_3 + u = C + D + E,$$

and adding $D + E$ gives the remaining plus sign. A linear functional taking values $(3, 1, -1, -3)$ on (f_1, f_2, f_3, f_4) and value 1 on u is strictly positive on all five generators and all ten displayed weights. Consequently none of their negatives lies in the color cone. The inversion sets under the four simple reflections are respectively

$$\begin{array}{c|l} \beta_1 & C, D \\ \beta_2 & A, B \\ \beta_3 & C, E \\ \eta & D, E \end{array}$$

where the first three reflections interchange adjacent f_i 's and the last changes the sign of u . They are exactly the virtual colors belonging to those roots, proving Statement 7.1.5.

The two type-T color orbits give $S_X = (\text{Std}_4 \oplus \text{Std}_4^*)[1]$. For the dual D_5 group its standard orthogonal module is $(V_1 \otimes U) \oplus W$, with $\dim U = 2$, $\dim W = 6$. Its exterior square gives

$$\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X = (U \otimes W)[3] \oplus \mathbf{1}[4].$$

Here $\mathfrak{so}(W) \simeq \mathfrak{sl}_4$, with $W = \bigwedge^2 \text{Std}_4$. Thus the six positive quotient weights and four positive S_X weights are exactly the ten entries just proved. All modules are interpreted on the actual central quotient of the dual group; the half-spin notation for the two 4's keeps this requirement visible, unlike writing them as representations of an unspecified SO_6 .

13 The exceptional simple ambient groups

We use Bourbaki numbering. In E_6, E_7, E_8 , the horizontal chain is 1, 3, 4, 5, 6, 7, 8, with vertex 2 attached to vertex 4. All groups in the inventory below mean the actual connected subgroup in the simply connected ambient group, rather than an arbitrary isogenous abstract product. A dot suppresses the finite central intersection.

Knop–Röhrle, *Spherical subgroups in simple algebraic groups*, arXiv:1305.3183v3, Table 2, reproduces the characteristic-zero exceptional part of Krämer's list. There are fourteen proper pairs:

G	H retained (no type N)	H excluded (type N)
G_2	A_2	$A_1 \cdot \tilde{A}_1$
F_4	B_4	$C_3 \cdot A_1$
E_6	$D_5, D_5 \cdot \mathbb{G}_m, F_4$	$C_4, A_5 \cdot A_1$
E_7	$E_6 \cdot \mathbb{G}_m$	$A_7, D_6 \cdot A_1$
E_8	none	$D_8, E_7 \cdot A_1$

The omitted improper pair G/G is a point and has trivial scattering. The spherical systems are [BP15, Section 3, cases 18–30 and 40]. The subgroup $D_5 \subset E_6$ is an augmentation of case 18, and must not be lost by listing only wonderful or spherically closed subgroups. Conversely, the disconnected normalizers in cases 23 and 41 are not additional connected pairs.

Here is an explicit type-N witness for every excluded row:

(G, H)	BP case	spherical root of type N
$(G_2, A_1 \tilde{A}_1)$	30	$2\alpha_1$
$(F_4, C_3 A_1)$	29	$2\alpha_1$
(E_6, C_4)	21	$2\alpha_1$
$(E_6, A_5 A_1)$	20	$2\alpha_2$
(E_7, A_7)	25	$2\alpha_1$
$(E_7, D_6 A_1)$	24	$2\alpha_1$
(E_8, D_8)	27	$2\alpha_1$
$(E_8, E_7 A_1)$	26	$2\alpha_7$

These are read from the single circles below vertices in the Luna diagrams, using [BP15, Table 1]. Passing from the normalizer to the connected A_7 in E_7 cannot halve $2\alpha_1$: the color belonging to $2\alpha_3$ evaluates on α_1 as $-1/2$, contradicting the integrality of the augmented Cartan pairing. The analogous issue for $C_4 \subset E_6$ cannot halve a root either: the extension obtained by removing $Z(E_6)$ has odd degree 3.

13.1 The spherical and weighted-color data

Put

$$\begin{aligned} a &= \alpha_1 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6, \\ b &= 2\alpha_2 + \alpha_3 + 2\alpha_4 + \alpha_5, \\ u &= 2\alpha_1 + \alpha_2 + 2\alpha_3 + 2\alpha_4 + \alpha_5, \\ v &= \alpha_2 + \alpha_3 + 2\alpha_4 + 2\alpha_5 + 2\alpha_6. \end{aligned}$$

The data for the six retained pairs are

(G, H)	S^p	normalized spherical roots	types of supports
(G_2, A_2)	$\{2\}$	$2\alpha_1 + \alpha_2$	G_2
(F_4, B_4)	$\{1, 2, 3\}$	$\alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4$	F_4
$(E_6, D_5\mathbb{G}_m)$	$\{3, 4, 5\}$	a, b	A_5, D_4
(E_6, D_5)	$\{3, 4, 5\}$	a, b	A_5, D_4
(E_6, F_4)	$\{2, 3, 4, 5\}$	u, v	D_5, D_5
$(E_7, E_6\mathbb{G}_m)$	$\{2, 3, 4, 5\}$	u, v, α_7	D_5, D_5, A_1

The center contributes an extra character direction in E_6/D_5 , so its spherical rank is 3 although its semisimple spherical root system has rank 2. All four E -type rows have classical supports and are covered by [Sak13, Theorem 6.3.2]. This statement concerns the support of each spherical root, not the type of the ambient group.

We now prove Statement 7.1.5 for all six rows, in the split arithmetic form. A notation $(\lambda; r)$ records the factor $1 - q^{-r}e^\lambda$; a repeated entry records multiplicity. The signs are all positive in this form. Weights of zero torus character are immaterial for a scattering quotient and are omitted throughout these lists.

G_2/A_2 . Let t be the fundamental weight of the dual SL_2 , so its positive root is $2t$. There is one color, with valuation t and exponent $5/2$. Indeed $S^p = \{\alpha_2\}$, the color is moved by α_1 , and $\langle \alpha_1^\vee, \rho - \alpha_2/2 \rangle = 5/2$. The extra virtual color has exponent $\langle \check{\rho}, 2\alpha_1 + \alpha_2 \rangle - 5/2 = 1/2$. Thus

$$\Theta^+ = \{(t; 5/2), (t; 1/2)\}.$$

Its positive cone is the ray generated by t . The only reflection takes t to $-t$ and inverts exactly the two displayed virtual colors. This proves both clauses of Statement 7.1.5.

F_4/B_4 . Again use a fundamental weight t of the dual SL_2 . In the standard orthonormal realization of F_4 , take

$$\alpha_1 = e_2 - e_3, \quad \alpha_2 = e_3 - e_4, \quad \alpha_3 = e_4, \quad \alpha_4 = (e_1 - e_2 - e_3 - e_4)/2.$$

The spherical root is e_1 . Here $\rho_{P(X)} = (11/2)e_1$, and the unique color is the restriction of $\alpha_4^\vee$, giving t and exponent $11/2$. The height of the spherical root is 8, so the other virtual exponent is $8 - 11/2 = 5/2$. Hence

$$\Theta^+ = \{(t; 11/2), (t; 5/2)\}.$$

The same one-dimensional argument proves Statement 7.1.5. Notice carefully that the second exponent uses $\langle \check{\rho}, \gamma \rangle$, not $\langle \rho, \gamma^\vee \rangle$.

$E_6/(D_5\mathbb{G}_m)$. The Cartan pairings of the three colors with (a, b) are

$$\check{v}_1 = \check{v}_6 = (1, -1), \quad \check{v}_2 = (-1, 2).$$

Their exponents are $5/2, 5/2, 3$. To check these directly, the Levi of $P(X)$ is A_3 on $\{3, 4, 5\}$, with $\rho_{A_3} = \frac{3}{2}\alpha_3 + 2\alpha_4 + \frac{3}{2}\alpha_5$. Pair $\rho - \rho_{A_3}$ with $\alpha_1^\vee, \alpha_6^\vee, \alpha_2^\vee$. Realize the spherical root system as C_2 , with simple roots $e_1 - e_2, 2e_2$. Then the color valuations are $(e_1 - e_2)/2$ twice, and e_2 once. Their cone is

$$C = \mathbb{R}_{\geq 0}(e_1 - e_2) + \mathbb{R}_{\geq 0}e_2.$$

The complete positive multiset is

$$\Theta^+ = \{((e_1 - e_2)/2; 5/2)^{[2]}, ((e_1 + e_2)/2; 5/2)^{[2]}, (e_1; 3), (e_2; 3)\}.$$

Every element of the signed-permutation orbit lies in C or $-C$. The first simple reflection swaps e_1, e_2 and inverts precisely the two copies of $(e_1 - e_2)/2$; the second changes the sign of e_2 and inverts precisely e_2 . These are exactly the virtual colors belonging to a, b .

E_6/D_5 . Use the character basis $(\omega_1, \omega_6, \omega_2)$ and its dual (x, y, z) . The three color valuations are x, y, z , so their cone is the nonnegative orthant. The spherical roots, in this basis, are $(1, 1, -1)$ and $(-1, -1, 2)$, and their coroots are $x + y, z$. Thus the simple reflections act by

	x	y	z
s_a	$-y$	$-x$	$x + y + z$
s_b	$x + z$	$y + z$	$-z$

and preserve the exponents in the following list:

$$\Theta^+ = \{(x; 5/2), (y; 5/2), (x + z; 5/2), (y + z; 5/2), (z; 3), (x + y + z; 3)\}.$$

Applying the displayed reflections shows that the full orbit multiset is $\Theta^+ \sqcup (-\Theta^+)$; there are no mixed-sign additional weights. The inversion sets are exactly $\{x, y\}$ for s_a and $\{z\}$ for s_b . This proves Statement 7.1.5 with the central direction retained. Setting $x = y = (e_1 - e_2)/2$ and $z = e_2$ recovers the preceding row.

E_6/F_4 . The valuations of the two colors are the two simple coroots of the spherical A_2 root system, and each has exponent 4. For example, the D_4 Levi has

$$\rho_{D_4} = 3\alpha_2 + 3\alpha_3 + 5\alpha_4 + 3\alpha_5,$$

so $\langle \alpha_1^\vee, \rho - \rho_{D_4} \rangle = 4$, and likewise for $\alpha_6^\vee$. Thus Θ^+ is precisely the three positive roots of an A_2 root system, all with exponent 4. The color cone is its positive root cone. Every root has one sign, and a simple reflection inverts exactly its own simple root. This proves Statement 7.1.5.

$E_7/(E_6\mathbb{G}_m)$. In the B_3 spherical root realization $u = e_1 - e_2$, $v = e_2 - e_3$, $\alpha_7 = e_3$, the color valuations are

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3, \quad e_3.$$

Their exponents are respectively $4, 4, 1/2, 1/2$. The first two follow from the same D_4 computation, and the last two from the two type-T colors at α_7 . The positive multiset is

$$\Theta^+ = \{(e_i - e_j; 4), (e_i + e_j; 4) : i < j\} \sqcup \{(e_i; 1/2)^{[2]} : 1 \leq i \leq 3\}.$$

The color cone is generated by $e_1 - e_2, e_2 - e_3, e_3$. It contains exactly one sign of each displayed orbit weight. The first two simple reflections interchange adjacent coordinates and invert only the corresponding difference; the last changes the sign of e_3 and inverts only its two copies. This proves Statement 7.1.5.

13.2 Independent calculation of the representation factors

The following computations establish the weight-and-degree comparison with Definition 4.5.5 without invoking the BZSV representation conjecture. Write $R[d]$ for a representation put in degree d ; its Euler exponent is $d/2$. A scalar representation is written $\mathbf{1}$. Finite central quotients of the displayed groups are understood; the named representations are taken on the actual dual group. In the D_5 row the spin representations carry opposite characters of the central torus.

(G, H)	S_X	$\check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X$
(G_2, A_2)	$\text{Std}_2[1]$	$\text{Std}_2[5] \oplus \mathbf{1}[4]$
(F_4, B_4)	0	$\text{Std}_2[5] \oplus \text{Std}_2[11] \oplus \mathbf{1}[4] \oplus \mathbf{1}[8] \oplus \mathbf{1}[12]$
$(E_6, D_5\mathbb{G}_m)$	0	$2\text{Spin}_5[5] \oplus \text{Vec}_5[6] \oplus \mathbf{1}[2] \oplus \mathbf{1}[4] \oplus \mathbf{1}[8]$
(E_6, D_5)	0	$(\text{Spin}_5 \otimes \chi)[5] \oplus (\text{Spin}_5 \otimes \chi^{-1})[5] \oplus \text{Vec}_5[6] \oplus \mathbf{1}[4] \oplus \mathbf{1}[8]$
(E_6, F_4)	0	$\text{Ad}_{A_2}[8] \oplus \mathbf{1}[4] \oplus \mathbf{1}[12]$
$(E_7, E_6\mathbb{G}_m)$	$2\text{Std}_6[1]$	$\Lambda_0^2 \text{Std}_6[8] \oplus \mathbf{1}[4] \oplus \mathbf{1}[12]$

Here Spin_5 has dimension 4 and Vec_5 has dimension 5. The group of Lie type C_3 acts on Std_6 . The A_2 -adjoint representation has dimension 8. Every nonzero positive weight in this table, with its degree divided by 2, is exactly one entry of the corresponding Θ^+ above.

For completeness, the branchings proving the table are as follows. For G_2 , use $A_1 \times A_1$ and $\mathfrak{g}_2 = (3, 1) \oplus (1, 3) \oplus (2, 4)$, with the second factor the Arthur $\mathfrak{sl}_2$. For F_4 , use $A_1 \times C_3$ and

$$\mathfrak{f}_4 = (3, 1) \oplus (1, 21) \oplus (2, 14_{\omega_3}).$$

Under the principal $\mathfrak{sl}_2$ of C_3 , $21 = V_2 \oplus V_6 \oplus V_{10}$ and $14_{\omega_3} = V_9 \oplus V_3$, where V_j has highest weight j . Indeed the standard 6 is V_5 , and $\Lambda^3 V_5 - V_5 = V_9 \oplus V_3$ by listing its weights. Taking highest-weight lines and adding 2 to their highest weights gives degrees 4, 8, 12 and 11, 5.

For the two E_6/D_5 rows, start with $\mathfrak{e}_6 = \mathfrak{so}_{10} \oplus \mathbf{1} \oplus 16_+ \oplus 16_-$. Under $\mathfrak{so}_6 \oplus \mathfrak{so}_4 \simeq \mathfrak{sl}_4 \oplus \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$,

$$45 = (15, 1, 1) \oplus (1, 3, 1) \oplus (1, 1, 3) \oplus (6, 2, 2).$$

The two half-spin modules decompose as $(4, 2, 1) \oplus (4^*, 1, 2)$ and its opposite-chirality counterpart. Principal $\mathfrak{sl}_2$ in $\mathfrak{sl}_4$ gives $4 = V_3$, $6 = V_4 \oplus V_0$, and $15 = V_2 \oplus V_4 \oplus V_6$. The V_0 terms centralizing this triple form $\mathfrak{so}_5$ together with the central torus. The V_4 highest-weight lines form its vector 5, and the V_3 lines form two spin 4's. Removing the dual Lie algebra removes the torus precisely for $H = D_5$.

For E_6/F_4 , decompose $\mathfrak{e}_6$ under D_4 and its two-dimensional centralizer torus as $28 \oplus 21$ plus two copies of each of the three triality 8's. Under principal $\mathfrak{sl}_2(D_4)$ every such 8 is $V_6 \oplus V_0$, and $28 = V_2 \oplus 2V_6 \oplus V_{10}$. The V_0 terms form the dual A_2 Lie algebra; the six nonzero V_6 weights plus its two zero weights form another adjoint representation. For $E_7/(E_6\mathbb{G}_m)$ use

$$\mathfrak{e}_7 = (28, 1, 1, 1) \oplus (1, 3, 1, 1) \oplus (1, 1, 3, 1) \oplus (1, 1, 1, 3) \oplus (8_v, 2, 2, 1) \oplus (8_s, 2, 1, 2) \oplus (8_c, 1, 2, 2).$$

The same principal D_4 calculation gives the dual $\mathfrak{sp}_6$ in highest weight 0, and its primitive exterior square 14 in highest weight 6, with scalar highest weights 2 and 10. The two even-sphere colors give two standard symplectic representations.

The scalar factors and the zero weights inside Vec_5 , Ad_{A_2} and $\Lambda_0^2 \text{Std}_6$ contribute only constants in the spectral parameter. Such constants cancel in $\beta(\chi)/\beta(w\chi)$. A claim of literal equality with a canonically normalized β would require specifying these constants separately.

13.3 The two exceptional rank-one functional equations

The missing local input for the split forms is now available in Sakellaridis, *Transfer operators and Hankel transforms: horospherical limits and quantization*, arXiv:2407.19500, Theorem 2.4.2 and

Example 2.4.7. This section uses its characteristic-zero local-field hypotheses, its canonical integral models and a conductor-zero additive character. Put $t = \check{\gamma}/2$, $z = e^t(\chi)$, and put

$$(r_1, r_2) = \begin{cases} (5/2, 1/2) & G_2/A_2, \\ (11/2, 5/2) & F_4/B_4. \end{cases}$$

The spectral Radon operator R_χ has source parameter χ^{-1} and target parameter χ . The cited theorem gives the actual scattering operator $\mathcal{S}_{w,\chi} = \mu_X(\chi)R_\chi$, with

$$\mu_X(\chi) = \gamma(\chi, t, 1 - r_1, \psi^{-1})\gamma(\chi, t, 1 - r_2, \psi)\gamma(\chi, -2t, 0, \psi).$$

For an unramified character with value v on the uniformizer, the conductor-zero Tate factor is $\gamma(v, s) = (1 - vq^{-s})/(1 - v^{-1}q^{s-1})$. Consequently

$$\mu_X(z) = \frac{1 - z^{-2}}{1 - q^{-1}z^2} \prod_{i=1}^2 \frac{1 - q^{r_i-1}z}{1 - q^{-r_i}z^{-1}}. \quad (1)$$

This is a proved functional equation, as opposed to a determination of an absolute square.

Here is an explicit normalization check. The Gindikin–Karpelevich scalar of R_χ on the normalized spherical vector is

$$c_X(z) = \frac{1 - q^{-1}z^2}{1 - z^2} \prod_{i=1}^2 \frac{1 - q^{-r_i}z}{1 - q^{r_i-1}z}. \quad (2)$$

One can verify this without a normalization convention being implicit. Write the coroots in simple-coroot coordinates, and let (m, a) denote the restriction of such a coroot to $\chi\delta_{L(X)}^{-1/2}$, namely $z^m q^a$. The positive coroots outside the Levi have the following multiset:

$$\begin{array}{l} G_2 \mid (1, -3/2), (1, -1/2), (1, 1/2), (1, 3/2), (2, 0) \\ F_4 \mid (1, \pm 9/2), (1, \pm 7/2), (1, \pm 5/2), 2(1, \pm 3/2), 2(1, \pm 1/2), (2, 0). \end{array}$$

For example, for G_2 a coroot $b_1\alpha_1^\vee + b_2\alpha_2^\vee$ has $m = b_1$ and $a = (-3b_1 + 2b_2)/2$. For F_4 it has $m = b_4$ and $a = (2b_1 + 2b_2 + 2b_3 - 9b_4)/2$. These formulas follow from $2\rho_{L(X)} = \alpha_2$ for G_2 and $2\rho_{L(X)} = 5\alpha_1 + 8\alpha_2 + 9\alpha_3$ for F_4 , respectively. Multiplying $(1 - q^{a-1}z^m)/(1 - q^a z^m)$ over the displayed multisets telescopes to (2). The G_2 string contributes $(1 - q^{-5/2}z)/(1 - q^{3/2}z)$; the two F_4 strings contribute $(1 - q^{-11/2}z)/(1 - q^{9/2}z)$ and $(1 - q^{-5/2}z)/(1 - q^{3/2}z)$. In both cases (2, 0) supplies the first factor of (2).

Multiplication of (1) and (2) gives

$$\mu_X(z)c_X(z) = \frac{1 - z^{-2}}{1 - z^2} \prod_i \frac{1 - q^{-r_i}z}{1 - q^{-r_i}z^{-1}} = \frac{\beta_X(z^{-1})}{\beta_X(z)}, \quad \beta_X(z) = \frac{1 - z^2}{(1 - q^{-r_1}z)(1 - q^{-r_2}z)}.$$

For completeness, the inverse here is required by the conventions, not an extra sign choice. The defining asymptotics in §2.2.1, diagram (6), of the cited paper say that the χ and χ^{-1} coefficients of a spherical eigenfunction are related by $a_\chi = \mathcal{S}_{w,\chi}a_{\chi^{-1}}$. On the spherical line this is $a_\chi = \mu_X(z)c_X(z)a_{\chi^{-1}}$. Normalizing the coefficient of χ to one, as in [Sak13, (6.2)] and the manuscript's (4.3.3), therefore makes $B_w(\chi) = a_{\chi^{-1}}/a_\chi$. Thus

$$B_w(\chi) = \frac{\beta_X(z)}{\beta_X(z^{-1})} = -z^2 \prod_i \frac{1 - q^{-r_i}z^{-1}}{1 - q^{-r_i}z}.$$

The positive graded weights already computed for V_X give exactly this β_X , up to spectral-parameter-independent factors. This completes the split-model calculation for both exceptional supports; it is not a consequence of [Sak13, Theorem 7.2.1] alone.

The theorem used in this section explicitly assumes that both G and H are split. In particular it does not, by itself, establish the formula for a form of G_2/A_2 whose stabilizer is a unitary group. That arithmetic qualification must be kept separate from the completed geometric and split-model calculations above.

13.4 The unramified nonsplit form of G_2/A_2

The following calculation also handles the quadratic unramified form, rather than invoking the split-stabilizer hypothesis of the preceding reference for that form. Let $\varepsilon = \eta(\varpi)$, where η is the unramified quadratic character determining the form; thus $\varepsilon = 1$ for SL_3 and $\varepsilon = -1$ for the unitary stabilizer. Write $\gamma = 2\alpha_1 + \alpha_2$, with α_1 short, and $z = e^{\check{\gamma}/2}(\chi)$.

There is a Brion path with the two successive U labels α_1, α_2 and terminal T label α_1 . Indeed $s_2 s_1 \gamma = \alpha_1$; the maximal-rank orbit graph consists of the three short-root lines. At its bottom the stabilizer is the torus in the short-root PGL_2 : it is split or the norm-one torus of the quadratic extension defining η , respectively. Here is explicit descent of the entire path. A representative n_1 of s_1 normalizes the long-root A_2 , since Weyl transformations preserve root length. As s_1 is outside $W(A_2)$, it represents the nontrivial element of $N_G(A_2)/A_2 = \mu_2$. Moreover $n_1 \in L_{\alpha_1} \subset P_{\alpha_1}$, so the resulting equivariant automorphism preserves $Y = H \backslash HP_{\alpha_1}$. On $Y/U_{P_{\alpha_1}} \simeq T \backslash \mathrm{PGL}_2$ it is the Weyl involution. Twisting by the nontrivial quadratic cocycle therefore gives the norm-one torus, exchanging its two geometric closed Borel orbits. Equivariance also preserves YP_{α_2} and X , so both U steps and their integration diagrams descend. On the long-root A_2 the same representative induces the outer automorphism, giving the unitary form of the stabilizer.

There is no character twist of the bottom torus in the integration along the U directions. In the split model, the three positive long roots of A_2 are $\alpha_2, 3\alpha_1 + \alpha_2, 3\alpha_1 + 2\alpha_2$. The two additional roots in the unipotent radical of P_{α_1} are $\alpha_1 + \alpha_2$ and $2\alpha_1 + \alpha_2$. Their pairings with $\alpha_1^\vee$ are $-1, +1$, so their determinant character on the derived torus is trivial. The same assertion descends to the nonsplit torus (and its absolute-value character is also necessarily trivial). The two U steps and their Haar normalizations are unchanged by this unramified twist.

For clarity, write $\gamma(v, s)$ for the conductor-zero unramified Tate factor. The terminal rank-one scalar relative to its Radon operator is

$$\mu_{T,\varepsilon}(z) = \gamma(z, 1/2)\gamma(\varepsilon z, 1/2)\gamma(z^{-2}, 0).$$

This follows directly from the split and nonsplit torus calculations of [Sak13, §§5.6–5.7, especially (5.6)], by applying the rank-one Gindikin–Karpelevich scalar $(1 - q^{-1}z^2)/(1 - z^2)$ and the inverse asymptotic-coefficient convention explained above. In particular, for $\varepsilon = -1$ the resulting normalized cocycle is $-z^2(1 - q^{-1}z^{-2})/(1 - q^{-1}z^2)$, exactly the nonsplit formula. This argument uses only unramified characters, so no ramified gamma-factor normalization has been tacitly asserted.

The conjugation of this equation along the two U steps is the formal Radon-inversion calculation in arXiv:2407.19500, Lemma 2.3.3 and the commutative diagram (33) in the proof of Theorem 2.4.2. That diagram is obtained by integration over the U orbits and holds equally with the nonsplit terminal torus; splitness of the terminal torus is needed only for the terminal scalar, which was just replaced by its proved nonsplit value. The two inversion contributions are

$$[\gamma(z, -3/2)\gamma(z^{-1}, 3/2)][\gamma(z, -1/2)\gamma(z^{-1}, 1/2)] = \frac{\gamma(z, -3/2)}{\gamma(z, 1/2)}.$$

Here the shifts can be checked from $\rho_{L(X)} = \alpha_2/2$:

$$\langle \alpha_1^\vee, \rho_{L(X)} \rangle = -3/2, \quad \langle s_1 \alpha_2^\vee, \rho_{L(X)} \rangle = -1/2, \quad \langle \alpha_1^\vee, s_2 s_1 \rho_{L(X)} \rangle = 0.$$

The last equality also checks that no shift was introduced into the terminal spectral parameter. We used the Tate identity $\gamma(z, s)\gamma(z^{-1}, 1 - s) = 1$ for unramified characters; the change from ψ to ψ^{-1} contributes no sign on such characters. Consequently

$$\mu_{G_2,\varepsilon}(z) = \gamma(z, -3/2)\gamma(\varepsilon z, 1/2)\gamma(z^{-2}, 0).$$

The ambient Gindikin–Karpelevich scalar is independent of the form of H . Multiplying by the G_2 scalar in (2) gives

$$\mu_{G_2,\varepsilon}(z)c_{G_2}(z) = \frac{1 - z^{-2}}{1 - z^2} \frac{1 - q^{-5/2}z}{1 - q^{-5/2}z^{-1}} \frac{1 - \varepsilon q^{-1/2}z}{1 - \varepsilon q^{-1/2}z^{-1}}.$$

Taking the inverse as before proves

$$B_w(\chi) = \frac{\beta_\varepsilon(z)}{\beta_\varepsilon(z^{-1})}, \quad \beta_\varepsilon(z) = \frac{1 - z^2}{(1 - q^{-5/2}z)(1 - \varepsilon q^{-1/2}z)}.$$

Finally this is the proposed L -factor with the Galois action required in Definition 4.5.3. The degree-five standard summand of $\check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X$ has sign $+1$. The degree-one standard summand S_X is generated by the difference of the two terminal geometric orbits; Frobenius acts on that difference by ε . Thus its positive weight has Euler factor $(1 - \varepsilon q^{-1/2}z)^{-1}$. This proves the unramified nonsplit quadratic case under the same integral-model and residue-characteristic hypotheses as the rank-one torus calculations used above.

13.5 Galois signs for the exceptional rows

For split G , Galois fixes the simple roots and the valuation lattice. A geometric color can be exchanged only with a second color moved by the same simple root and having the same valuation. In particular, equal valuations alone do not permit an exchange between colors moved by different simple roots. The graded quotient $Q_X = \check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X$ is obtained from the split pinned adjoint representation. Its weight spaces in the calculations above have the split Frobenius action; the additional signs to be checked are those on S_X in Definition 4.5.3.

For F_4/B_4 , $E_6/(D_5\mathbb{G}_m)$, E_6/D_5 and E_6/F_4 one has $S_X = 0$, so there are no degree-one signs. In the first two cases the stabilizer is self-normalizing. In the third its equivariant automorphism group is a split one-dimensional torus; its first Galois cohomology is trivial. For simply connected E_6/F_4 the equivariant automorphisms are its central μ_3 ; central twists do not permute colors or change the adjoint A_2 weights computed above. Such central-isogeny orbits must still be interpreted using the actual character lattice and the integral model, rather than silently replacing a connected stabilizer by its normalizer.

For G_2/A_2 the normalizer quotient is μ_2 . The short-root Weyl representative normalizes the long-root A_2 and inverts the terminal torus in L_{α_1} . The nontrivial quadratic form therefore exchanges the two terminal geometric orbits. Their difference, used by Definition 4.5.3, has sign -1 . Hence

$$S_X = \text{Std}_2[1] \otimes \eta, \quad Q_X = \text{Std}_2[5] \oplus \mathbf{1}[4],$$

where η is the associated unramified quadratic character. The preceding explicit computation proves the scattering factor with this sign.

For $E_7/(E_6\mathbb{G}_m)$ the normalizer quotient is μ_2 . Its nontrivial element exchanges the two colors moved by α_7 , whose common valuation is e_3 . Frobenius consequently acts by either the identity or the transposition on the two standard summands of S_X . For the nonsplit form their eigenvalues are $+1, -1$, giving

$$S_X = \text{Std}_6[1] \otimes (\mathbf{1} \oplus \eta), \quad Q_X = \bigwedge_0^2 \text{Std}_6[8] \oplus \mathbf{1}[4] \oplus \mathbf{1}[12].$$

This exactly gives the two virtual signs in [Sak13, §7.1]. The spherical root α_7 is classical, so its nonsplit torus functional equation is already included in [Sak13, Theorem 6.3.2]. The signed versions of the cone and inversion proofs require no new step: the Weyl group acts on the weight coordinate and preserves the sign and degree.

The four classical ambient exceptional-stabilizer cases have $S_X = 0$ except $\text{Spin}_{10}/(\text{SO}_2 \text{Spin}_7)$. In that case the two even-sphere colors have the different valuations f_2 and $-f_3$, so they cannot be exchanged when G is split. Thus its $\text{Std}_4 \oplus \text{Std}_4^*$ has split Frobenius and introduces no negative virtual sign.

14 The symplectic product families: explicit verification

Put $a = m + 1$, $b = n + 1$, and, in the three-arm family, $c = \ell + 1$. We use the usual coordinates for C_a : $\alpha_i = e_i - e_{i+1}$ ($i < a$), $\alpha_a = 2e_a$, and $\omega_j = e_1 + \cdots + e_j$. For an arm with $a \geq 2$ put

$$\eta = e_1 - e_2, \quad \tau = e_1 + e_2 = \alpha_1 + 2\alpha_2 + \cdots + 2\alpha_{a-1} + \alpha_a.$$

These are orthogonal spherical roots. Write x, y for the corresponding fundamental weights of the dual $A_1 \times A_1$, so that $\langle x, \eta \rangle = \langle y, \tau \rangle = 1$ and the other two pairings vanish. For $a = 1$, only $\eta = \alpha_1$ and x are present. The product-family spherical roots used below are the ones in Bravi–Pezzini, *The spherical systems of the wonderful reductive subgroups*, cases 42, 43, 47, and 48. The color weights follow independently from the free weight monoids in Avdeev, arXiv:1012.0132, Table 1, rows 5–8. Thus the checks below specify the color cone, all Weyl orbits, and the crossing sets; they do not assume Statement 7.1.5.

A normalization that matters. In Sakellaridis’s definition of the companion virtual color its exponent is

$$r' = \langle \check{\rho}, \gamma \rangle - r,$$

where $\check{\rho}$ is the half-sum of positive *coroots*. It is not $\langle \rho, \check{\gamma} \rangle - r$. For $\gamma = \tau$ this height is $2a - 2$. This distinction is essential for all the symplectic arms.

14.1 Case 3: the diagonal triple SL_2

Let x_1, x_2, x_3 be the three fundamental dual weights. The free monoid of regular highest weights is generated by $\omega_1 + \omega_2$, $\omega_1 + \omega_3$, and $\omega_2 + \omega_3$. Its dual color valuations are

$$c_{12} = x_1 + x_2 - x_3, \quad c_{13} = x_1 - x_2 + x_3, \quad c_{23} = -x_1 + x_2 + x_3.$$

All three have exponent $1/2$. Their Weyl orbit is the eight weights $\pm x_1 \pm x_2 \pm x_3$. Exactly the four weights with at least two plus signs are in the color cone: the three displayed generators and their sum $x_1 + x_2 + x_3$. Every other weight is the negative of one of these four. The reflection changing the sign of x_i crosses precisely the two colors c_{ij} and c_{ik} . This proves both assertions of Statement 7.1.5. Here $S_X = \mathbf{2} \otimes \mathbf{2} \otimes \mathbf{2}$ in degree one, $\check{\mathfrak{g}}_X = \mathfrak{sl}_2^3$, and $\check{\mathfrak{g}}^e/\check{\mathfrak{g}}_X = 0$. These statements use the exact character lattice generated by the three pair sums; the dual group is the corresponding central cover of the ambient dual group, and its distinguished morphism need not be injective on centers.

14.2 Cases 4(a) and 5: two symplectic arms

First suppose $a, b \geq 2$. The spherical roots are τ_A, τ_B and $\gamma_0 = \alpha_1^A + \alpha_1^B$; their dual root system is A_1^3 . Denote the three fundamental dual weights by y_A, y_B, x . The highest-weight monoid has the free generators

$$\lambda_A = \omega_2^A, \quad \lambda_B = \omega_2^B, \quad \mu = \omega_1^A + \omega_1^B.$$

The colors are $y_A - x$, $y_B - x$, and $2x$, with respective exponents $a - 1$, $b - 1$, and 1. The complete virtual Weyl multiset is

$$\{\pm y_A \pm x\}_{a-1} \sqcup \{\pm y_B \pm x\}_{b-1} \sqcup \{\pm 2x\}_1, \tag{P1}$$

where a subscript is the Euler-factor exponent r , not the grading degree. The positive weights are $y_A \pm x$, $y_B \pm x$, and $2x$. For example, $y_A + x = (y_A - x) + 2x$ is in the color cone. Conversely, a weight with negative y_A -coefficient cannot be in that cone; this proves the positive/negative partition, including the same assertion for the other arm. Changing the sign of y_A crosses exactly

$y_A - x, y_A + x$; the second is its virtual companion with the same exponent $a - 1$. Changing the sign of x crosses only $2x$, and interchanges each pair $y_A \pm x, y_B \pm x$. This proves Statement 7.1.5, with multiplicities and exponents, for case 5.

An arm of size $a = 1$ contributes no y_A and no corresponding four-weight orbit. Deleting it gives case 4(a). Deleting both leaves the diagonal group case. The remaining orbit $\{\pm 2x\}_1$ persists in all these specializations. There are no even-sphere colors in this two-arm family, so $S_X = 0$. Modulo zero-weight summands, the graded quotient is

$$(\text{Ad } \mathbf{2}_x)[2] \oplus (\mathbf{2}_x \otimes \mathbf{2}_{y_A})[2a - 2] \oplus (\mathbf{2}_x \otimes \mathbf{2}_{y_B})[2b - 2], \quad (\text{P2})$$

with absent arms omitted. The notation $[d]$ means degree d . The positive nonzero weights of (P2) are exactly (P1)'s positive part, with $r = d/2$.

14.3 Case 7: three symplectic arms

The free highest-weight generators are the three pair sums $\mu_{ij} = \omega_1^i + \omega_1^j$ and, for each arm $a_i \geq 2$, $\lambda_i = \omega_2^i$. The colors are

$$c_{ij} = x_i + x_j - x_k \quad (\{i, j, k\} = \{1, 2, 3\}), \quad d_i = y_i - x_i.$$

The first three have exponent $1/2$, and d_i has exponent $a_i - 1$. The complete virtual Weyl multiset is

$$\{\pm x_1 \pm x_2 \pm x_3\}_{1/2} \sqcup \bigsqcup_{a_i \geq 2} \{\pm y_i \pm x_i\}_{a_i - 1}. \quad (\text{P3})$$

The positive part consists of the four majority-plus triple weights and, for each arm, $y_i \pm x_i$. To check this directly, note that a triple weight has zero y -coordinates, so membership in the full color cone is exactly membership in the cone generated by c_{12}, c_{13}, c_{23} . Also $2x_i = c_{ij} + c_{ik}$, hence $y_i + x_i = d_i + c_{ij} + c_{ik}$ is positive. A negative y_i -coefficient precludes positivity. The reflections in x_i cross exactly c_{ij}, c_{ik} and exchange the two positive arm weights; the reflection in y_i crosses exactly $y_i - x_i, y_i + x_i$. This proves Statement 7.1.5 for every triple of positive integers (a_1, a_2, a_3) . Here

$$S_X = (\mathbf{2}_{x_1} \otimes \mathbf{2}_{x_2} \otimes \mathbf{2}_{x_3})[1], \quad (\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X)_{\neq 0} = \bigoplus_{a_i \geq 2} (\mathbf{2}_{x_i} \otimes \mathbf{2}_{y_i})[2a_i - 2].$$

In this display the subscript $\neq 0$ records the multiset of nonzero torus weights; it is not a separate canonical subrepresentation.

14.4 Case 6: a central Sp_4 and two arms

The dual semisimple Lie algebra is $\mathfrak{so}_5 \oplus \mathfrak{sl}_2^2$, with one further $\mathfrak{sl}_2$ for each nontrivial tail. Write (t_1, t_2, u, v) for a weight on the core, and measure these coordinates in units of $1/2$. Its four colors, in these units, are

$$c_0 = (1, 1, -1, -1), \quad c_A = (1, -1, 1, -1), \quad c_B = (1, -1, -1, 1), \quad c_* = (-1, 1, 1, 1). \quad (\text{P4})$$

They are the duals of ω_2^0 , $\omega_1^0 + \omega_1^A$, $\omega_1^0 + \omega_1^B$, and $\omega_2^0 + \omega_1^A + \omega_1^B$. Each has exponent $1/2$. Their Weyl orbit is the sixteen sign vectors $(t_1, t_2, u, v) \in \{\pm 1\}^4$. The coefficients of such a vector in the color basis (P4) are

$$\frac{1}{2}(t_1 + t_2), \quad \frac{1}{2}(t_1 + u), \quad \frac{1}{2}(t_1 + v), \quad \frac{1}{2}(t_1 + t_2 + u + v). \quad (\text{P5})$$

Consequently the eight positive weights are those with $t_1 = 1$ except $(1, -1, -1, -1)$, together with $(-1, 1, 1, 1)$. The remaining eight are their negatives. The four simple reflections and their crossing

sets are

$$\begin{array}{l|l} (t_1, t_2) \mapsto (t_2, t_1) & c_A, c_B \\ t_2 \mapsto -t_2 & c_0, c_* \\ u \mapsto -u & c_A, c_* \\ v \mapsto -v & c_B, c_* \end{array} \quad (\text{P6})$$

This follows by substitution into (P5), and is the required check for the core. Each tail attaches $y_i - x_i$ and the four-weight orbit $\{\pm y_i \pm x_i\}_{a_i-1}$. As in case 7, $2x_A = c_A + c_*$ and $2x_B = c_B + c_*$, so the attachment preserves the partition and adds precisely the two expected crossings for the new tail reflection. Thus Statement 7.1.5 is proved for case 6, including $a = 1$ or $b = 1$. The representation is

$$S_X = (\text{Spin}_5 \otimes \mathbf{2}_{x_A} \otimes \mathbf{2}_{x_B})[1],$$

where Spin_5 denotes the four-dimensional spin representation, and the nonzero quotient weights are the two arm representations in degrees $2a - 2$ and $2b - 2$, with absent tails omitted.

14.5 Case 4(b): the diagonal Sp_4 family

Write the pair as $G = \text{Sp}_{2a} \times \text{Sp}_4$, $H = \text{Sp}_{2a-4} \times \text{Sp}_4$, with $a \geq 3$. For $a \geq 4$ the spherical roots on the first factor are

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_3 + e_4,$$

and on the second they are $f_1 - f_2, 2f_2$. Hence the dual root system is $D_4 \times B_2$. The following six highest weights freely generate the monoid (Avdeev, Table 1, row 6):

$$\lambda_1 = \omega_1 + \varpi_1, \quad \lambda_2 = \omega_2 + \varpi_2, \quad \lambda_3 = \omega_3 + \varpi_1, \quad \lambda_4 = \omega_4, \quad \lambda_5 = \omega_2, \quad \lambda_6 = \omega_1 + \omega_3 + \varpi_2. \quad (\text{P7})$$

In dual coordinates $(E_1, E_2, E_3, E_4, F_1, F_2)$ the color valuations are

	E_1	E_2	E_3	E_4	F_1	F_2
$2c_1$	1	-1	-1	1	1	-1
$2c_2$	-1	1	-1	1	1	1
$2c_3$	-1	1	1	-1	1	-1
c_4	0	0	0	1	0	0
$2c_5$	1	1	-1	-1	-1	-1
$2c_6$	1	-1	1	-1	-1	1

(P8)

The five half-integral colors have exponent $1/2$. Their common Weyl orbit is

$$\frac{1}{2}(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4, \delta_1, \delta_2), \quad \epsilon_i, \delta_j \in \{\pm 1\}, \quad \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 = 1. \quad (\text{P9})$$

The color $c_4 = E_4$ has exponent $a - 3$; its orbit is $\{\pm E_i\}_{1 \leq i \leq 4}$, and its companion at $e_3 + e_4$ is E_3 , with the same exponent.

Here is a complete sign check for the 32 weights (P9). For each listed ϵ , the table gives exactly the allowed δ for positivity; all unlisted choices are negatives of the listed weights.

$\epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4$	$\delta_1 \delta_2$
- + - +	++
- + + -	+ -, ++
+ - - +	+ -, ++
+ - + -	- +, + -, ++
+ + - -	--, - +, + -, ++
+ + + +	--, - +, + -, ++

(P10)

For verification, the six color-basis coefficients of a vector $w = (t_1, t_2, t_3, t_4, u, v)$ are simply

$$t_1 + u, \quad t_1 + t_2 + u + v, \quad t_1 + t_2 + t_3 + u, \quad t_1 + t_2 + t_3 + t_4, \quad t_1 + t_2, \quad 2t_1 + t_2 + t_3 + u + v. \quad (\text{P11})$$

Substituting the sign vectors of (P9) into these six inequalities proves (P10) without any geometric assumption. The six simple-reflection crossing sets on the spin orbit are, in the order of roots just given,

$$\{c_1, c_6\}, \quad \{c_2, c_5\}, \quad \{c_3, c_6\}, \quad \emptyset, \quad \{c_1, c_3\}, \quad \{c_2, c_6\}. \quad (\text{P12})$$

On the vector orbit the positive weights are E_1, E_2, E_3, E_4 ; only $e_3 + e_4$ crosses weights, namely E_3, E_4 . Equations (P10)–(P12) prove Statement 7.1.5 completely. The graded representation has

$$S_X = (\text{Spin}_8^+ \otimes \text{Spin}_5)[1], \quad (\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X)_{\neq 0} = \text{Std}_8[2a - 6]. \quad (\text{P13})$$

The choice of half-spin is fixed by the even-sign convention in (P9).

When $a = 3$, delete λ_4 and E_4 from (P7)–(P8). The dual root system is $B_3 \times B_2$, and S_X is $\text{Spin}_7 \otimes \text{Spin}_5$, again of dimension 32. Its weights are all $\frac{1}{2}(\epsilon_1, \epsilon_2, \epsilon_3, \delta_1, \delta_2)$. For the positivity test extend them by $\epsilon_4 = \epsilon_1 \epsilon_2 \epsilon_3$ and use (P10). The crossing sets for $e_1 - e_2, e_2 - e_3, 2e_3, f_1 - f_2, 2f_2$ are

$$\{c_1, c_6\}, \quad \{c_2, c_5\}, \quad \{c_3, c_6\}, \quad \{c_1, c_3\}, \quad \{c_2, c_6\}.$$

There is no vector-orbit quotient in this case. At $a = 2$ the pair is the diagonal Sp_4 group case and should be treated there.

14.6 Why the Lie-algebra quotient gives the displayed Euler weights

For a nontrivial symplectic arm the standard representation of the ambient dual group SO_{2a+1} restricts to

$$W \oplus \text{Sym}^{2a-4} \mathbf{2}, \quad W = \mathbf{2}_x \otimes \mathbf{2}_y.$$

The second factor carries the distinguished Arthur SL_2 . In $\mathfrak{so}_{2a+1} = \bigwedge^2(W \oplus \text{Sym}^{2a-4} \mathbf{2})$, the term $\bigwedge^2 W$ is $\mathfrak{sl}_{2,x} \oplus \mathfrak{sl}_{2,y}$. The mixed term has highest Arthur weight $2a - 4$, so its e -centralizer contributes W in degree $2a - 2$. The remaining terms are trivial $\check{G}_X$ -modules $\mathbf{1}[4j]$ for $1 \leq j \leq a - 2$: these come from the principal exponents $1, 3, \dots, 2a - 5$ of $\mathfrak{so}_{2a-3}$. In the two-arm case the two copies of $\mathfrak{sl}_{2,x}$ are divided by their diagonal, leaving the adjoint summand in degree two. For case 4(b) the analogous decomposition is $\text{Std}_{2a+1} = \text{Std}_8 \oplus \text{Sym}^{2a-8} \mathbf{2}$, giving degree $2a - 6$. In the latter case the additional trivial modules are $\mathbf{1}[4j]$, $1 \leq j \leq a - 4$. The positive nonzero weights are those selected above: $y \pm x, 2x$, or $E_1, \dots, E_4$. Thus they match the Euler factors of $\check{\mathfrak{g}}^e / \check{\mathfrak{g}}_X$ directly, not just up to an unspecified conjectural representation. If the positive quotient means $p(\check{\mathfrak{g}}^e \cap \check{\mathfrak{n}}^+)$, the highest vectors in these trivial modules are also included. They multiply β by $\prod_j (1 - q^{-2j})^{-1}$ for each applicable tail. Such constants cancel in $\beta(\chi) / \beta(w\chi)$. The zero-weight Cartan vector in the extra adjoint module of the two-arm case is not in the positive root subalgebra.

All normalized spherical roots in these families are ordinary simple roots, $a_1 \times a_1$ roots, or the indicated c_a roots. None is a type- N root. Sakellaridis's Statement 6.3.1 applies by his Theorem 6.3.2 because all support Levi factors here are classical. Subject to the arithmetic and integral-model hypotheses of the analytic formula, the explicit checks above therefore establish the required cocycle formula for these families.

Arithmetic scope. The geometric checks must not be confused with local multiplicity one. For cases 3, 4(b), 6, and 7 in their simply connected forms the highest-weight lattice has index two in its saturation inside the ambient weight lattice. For case 3 this is the even-total-parity lattice; for 4(b) the matrix of (P7) in fundamental weights has determinant -2 (also after deleting ω_4 at $a = 3$); the four core generators in case 6 have determinant ± 2 ; case 7 has the same parity core as case 3. Adding the tail generators does not change this index. Thus the geometric generic Borel stabilizer has a μ_2 component. For the standard split models, this also proves failure of Assumption (4.5.1), not merely a potential difficulty. Indeed choose a rational point in the geometric open Borel orbit; this is possible because its preimage in split G is nonempty and $G(k)$ is Zariski dense. The

image D of its stabilizer B_x in the split torus has character group $X^*(T)/\mathcal{X}(X) \simeq \mathbb{Z}^d \oplus \mathbb{Z}/2$, and hence $D \simeq \mathbb{G}_m^d \times \mu_2$. The kernel is unipotent. Since the field has characteristic zero,

$$H^1(k, B_x) \simeq H^1(k, D) \simeq k^\times / (k^\times)^2, \quad H^1(k, B) = 1.$$

Consequently the geometric open orbit has $|k^\times / (k^\times)^2| > 1$ rational $B(k)$ -orbits. Sakellaridis, Theorem 3.2.2, identifies the generic unramified multiplicity with g times this number, where $g \geq 1$ is the generic degree of the Satake-parameter map. It follows that the exact split simply connected cases 3, 4(b), 6, and 7 are outside the local multiplicity-one hypothesis of the manuscript's Theorem 4.5.8, even though the full geometric and Euler-factor checks above are valid. Changing the central form can change this arithmetic conclusion.

There is a different obstruction in case 5: Sakellaridis's Example 8.0.1 is exactly $\mathrm{SL}_2^3 \backslash (\mathrm{Sp}_4 \times \mathrm{Sp}_4)$, namely $a = b = 2$. The discussion after Corollary 8.0.4 shows that this example has generic multiplicity one but fails multiplicity one at some unramified parameters. Hence even saturation of the weight lattice does not establish the manuscript's hypothesis for all irreducible unramified representations.

15 The orthogonal Gross–Prasad family and its triality form

This section treats Brion cases 4(c) and 4(d) at the level of the geometric root datum and the positive Euler factor. Their rational multiplicity-one scope is recorded separately below. Root-coordinate computations are unchanged by central isogeny, but the character lattice and the number of rational open Borel orbits need not be unchanged.

Geometric input. For the orthogonal pair

$$\mathrm{diag}(\mathrm{SO}_d) \subset \mathrm{SO}_d \times \mathrm{SO}_{d+1},$$

the spherical systems in BP cases 46 and 50 have $S^p = \emptyset$, and every simple root of the ambient group is a type-T spherical root. Thus the dual Lie algebra equals the ambient dual Lie algebra, the Arthur $\mathfrak{sl}_2$ is trivial, and the centralizer-quotient part of V_X is zero. The color valuations in the following calculation are the two type-T colors at each simple root; their pairings with the simple roots reproduce these BP diagrams. In the customary effective orthogonal form their dual root system is $C_m \times D_n$, with $m = n$ or $m = n - 1$. Write $e_1, \dots, e_m$ for the standard C_m coordinates and $f_1, \dots, f_n$ for the standard D_n coordinates. The colors all lie in the single Weyl orbit

$$\Theta = \{\epsilon e_i + \epsilon' f_j : \epsilon, \epsilon' \in \{1, -1\}\},$$

with multiplicity one. The dominant member of this orbit is the sum of the first fundamental weights of the two factors. Both fundamental representations are minuscule; hence the representation specified by the highest-weight construction is

$$S_X = \mathrm{Std}_{\mathrm{Sp}_{2m}} \otimes \mathrm{Std}_{\mathrm{SO}_{2n}}, \quad V_X = S_X,$$

entirely in degree one. Its weights are exactly Θ , so there is no unresolved comparison between interior representation weights and a Weyl orbit in this family. For a different central form this description is interpreted on its actual dual torus or finite covering torus, on which the displayed weights are characters.

An explicit cone certificate. For $m = n$, order the coordinates as

$$z_1 = e_1, z_2 = f_1, z_3 = e_2, z_4 = f_2, \dots, z_{2n-1} = e_n, z_{2n} = f_n.$$

For $m = n - 1$, order them as

$$z_1 = f_1, z_2 = e_1, z_3 = f_2, z_4 = e_2, \dots, z_{2n-2} = e_{n-1}, z_{2n-1} = f_n.$$

Put $N = m + n$, and define

$$\delta_i = z_i - z_{i+1} \quad (1 \leq i < N), \quad \delta_N = z_{N-1} + z_N, \quad C = \sum_{i=1}^N \mathbf{R}_{\geq 0} \delta_i.$$

The union of the color valuations displayed below is precisely $\{\delta_1, \dots, \delta_N\}$, with repetitions when a color is moved by more than one simple root. In particular C is the color cone. It is a positive-root cone of type D_N . Each occurring weight is a root $\pm z_a \pm z_b$ of that larger root system, for $a < b$. Its positive choice is $z_a - z_b$ or $z_a + z_b$. Explicit certificates are

$$z_a - z_b = \delta_a + \dots + \delta_{b-1},$$

and, for $b \leq N - 1$,

$$z_a + z_b = \delta_a + \dots + \delta_{b-1} + 2(\delta_b + \dots + \delta_{N-2}) + \delta_{N-1} + \delta_N,$$

while

$$z_a + z_N = \delta_a + \dots + \delta_{N-2} + \delta_N \quad (a < N).$$

Empty sums are zero. These identities prove that every $\theta \in \Theta$ belongs to C or $-C$. Choose any functional with $\ell(z_1) > \dots > \ell(z_N) > 0$; it is strictly positive on every δ_i , so the signs are disjoint and agree with this ordering. This proves the first assertion of Statement 7.1.5 without a numerical rank restriction.

Complete crossing list when $m = n$. Use the simple roots $a_i = e_i - e_{i+1}$ for $i < n$, $a_n = 2e_n$, $b_j = f_j - f_{j+1}$ for $j < n$, and $b_n = f_{n-1} + f_n$. The positive weights changed to negative ones are

reflection	crossing weights	
$s_{a_i}, i < n$	$e_i - f_i, f_i - e_{i+1}$	(GP1)
s_{a_n}	$e_n - f_n, e_n + f_n$	
$s_{b_j}, j < n$	$f_j - e_{j+1}, e_{j+1} - f_{j+1}$	
s_{b_n}	$f_{n-1} - e_n, e_n + f_n$	

Each row is precisely the pair of color valuations for that simple root. To verify that the list is exhaustive, a reflection of the first or third kind exchanges two coordinates of one factor. Only the unique intervening coordinate of the other factor can change which summand has larger absolute order in $\epsilon e_i + \epsilon' f_j$. All other weights retain their sign. The reflection s_{a_n} changes the sign of the last e -coordinate and only meets the smaller f_n . Finally s_{b_n} sends f_{n-1} to $-f_n$ and f_n to $-f_{n-1}$; only e_n , lying between them, produces the two listed crossings. These observations give the displayed weights by direct substitution, and exhaust all four sign choices for every pair.

Complete crossing list when $m = n - 1$. Now use $a_i = e_i - e_{i+1}$ for $i < m$, $a_m = 2e_m$, and the same D_n simple roots. The answer is

reflection	crossing weights	
$s_{a_i}, i < m$	$e_i - f_{i+1}, f_{i+1} - e_{i+1}$	(GP2)
s_{a_m}	$e_m - f_n, e_m + f_n$	
$s_{b_j}, j < n$	$f_j - e_j, e_j - f_{j+1}$	
s_{b_n}	$f_{n-1} - e_m, e_m + f_n$	

The same coordinate-exchange argument proves completeness. In each row the two weights sum to the simple root labeling that row and are sent to the negatives of one another. Their grades are one, so their weighted exponents are both $r = 1/2$. Thus (GP1) and (GP2) prove exactly the second assertion of Statement 7.1.5, with all multiplicities and gradings. These formulas also cover the semisimple low-rank case $D_2 = A_1 \times A_1$.

The resulting beta-function. Writing Θ^+ for the weights selected by the indicated order, we obtain

$$\beta(t) = \frac{\prod_{\alpha \in R^+(C_m \times D_n)} (1 - \alpha(t))}{\prod_{\theta \in \Theta^+} (1 - q^{-1/2} \theta(t))}.$$

Every ambient simple spherical root has support A_1 , so its analytic rank-one factor is covered by Sakellaridis's Theorem 6.3.2. Applying the reflection-crossing lemma to (GP1) or (GP2), followed by the cocycle relation, proves the asserted beta cocycle for this geometric family in the split form. This proves the required combinatorial hypotheses and the V_X^+ comparison for the family; whether a selected rational central form meets Assumption 4.5.1 is a separate question.

Triality: Brion case 4(d). Over the simply connected group Spin_8 , triality carries the spinor Spin_7 subgroup to the vector stabilizer Spin_7 . Therefore the pair $\mathrm{diag}(\mathrm{Spin}_7) \subset \mathrm{Spin}_7 \times \mathrm{Spin}_8$, with the spinor embedding in the second factor, is carried to the $d = 7$ orthogonal Gross–Prasad pair. Triality transports the torus lattice, simple roots, colors, positive cone, and grading, so it transports the entire preceding certificate. This proves the geometric and Euler-factor assertion for case 4(d). Triality permutes the three central quotients of Spin_8 ; it need not descend to a prescribed quotient called SO_8 . A statement about that quotient must retain this central-form qualification.

16 A necessary arithmetic lattice test

For split G , let A denote the torus quotient of the Borel and let $A_{\mathrm{stab}} \subset A$ be the image of the stabilizer of a point in the open Borel orbit. It is a split diagonalizable group, and

$$0 \longrightarrow \mathcal{X}(X) \longrightarrow X^*(A) \longrightarrow X^*(A_{\mathrm{stab}}) \longrightarrow 0.$$

If $\mathcal{X}(X)$ is not saturated, the last group has a nontrivial finite cyclic direct summand $\mathbf{Z}/d\mathbf{Z}$. Accordingly A_{stab} has a factor μ_d , and

$$H^1(k, A_{\mathrm{stab}}) \supset H^1(k, \mu_d) = k^\times / (k^\times)^d \neq 1.$$

The nontriviality follows already from valuation modulo d . Sakellaridis's generic multiplicity theorem ([arXiv:math/0606130v2](#), Theorem 1.2.1) gives

$$m_{\mathrm{gen}}(X) = [N_W(\delta^{-1/2} A_X^*) : W_X] |H^1(k, A_{\mathrm{stab}})|.$$

Hence nonsaturation disproves Assumption 4.5.1 for the full space $X(k)$, regardless of the combinatorial beta identity. Saturation is only a necessary test here: the Weyl-normalizer index and nongeneric parameters still need consideration for an all-representations multiplicity-one assertion.

For the literal simply connected Gross–Prasad pair

$$\mathrm{diag}(\mathrm{Spin}_d) \subset \mathrm{Spin}_d \times \mathrm{Spin}_{d+1}, \quad d \geq 3,$$

the generic Borel stabilizer is finite. Indeed $\dim B_G = \dim X = d(d+1)/2$. It contains the nontrivial central element $(-1, -1)$, which is in the diagonal stabilizer and fixes every point. Thus its diagonalizable image is nontrivial finite, the above lattice is nonsaturated, and generic multiplicity is greater than one. The same observation applies to the triality form. These simply connected rows are therefore outside Assumption 4.5.1 as literally stated. Effective orthogonal quotients can have different multiplicity behavior; they cannot be interchanged with the simply connected pair without checking the arithmetic hypotheses.

More explicitly, in the triality form let z_{spin} be the central element of Spin_8 that is the image of -1 in the spinor Spin_7 subgroup. The central element in the generic stabilizer is $(-1, z_{\mathrm{spin}})$; it need not be $(-1, -1)$ in the vector-cover convention. In the literal group printed in the manuscript, $\mathrm{Spin}_7 \times \mathrm{SO}_8$, the spin representation is faithful and its center maps to $-I_8$. The

stabilizer consequently contains $(-1, -I_8)$, so the same finite Borel-stabilizer obstruction applies even to that particular non-simply connected ambient form. The familiar effective orthogonal pair $\mathrm{SO}_7 \subset \mathrm{SO}_7 \times \mathrm{SO}_8$ removes a different subgroup of the product center. Its multiplicity-one results must not be transferred to the printed triality form without tracking those quotients.

17 The two mixed type-A product families

17.1 Case 8: adjacent special-linear degrees

Let $G = \mathrm{SL}_n \times \mathrm{SL}_{n+1}$ and let H be the image of $\mathrm{SL}_n \times \mathbb{G}_m$ specified in Theorem 1.1.28. For $n \geq 2$, Avdeev's extended weight monoid has the $2n$ free generators

$$(\pi_{n-i} + \varphi_{i+1}, n-i) \quad (0 \leq i < n), \quad (\pi_{n-i} + \varphi_i, -i) \quad (1 \leq i \leq n), \quad (\mathrm{M1})$$

where $\pi_0 = \pi_n = 0$ and the second coordinate is measured in a primitive character of H/H^{der} . Use coordinates $u_1, \dots, u_n$ and $v_1, \dots, v_{n+1}$ with $\sum u_i = \sum v_j = 0$, and reverse the ordering of the first ambient simple-root system if necessary. The color valuations dual to (M1) are

$$a_i = v_i - u_i, \quad b_i = u_i - v_{i+1} \quad (1 \leq i \leq n). \quad (\mathrm{M2})$$

Their single relation is

$$\sum_{i=1}^n (n-i+1)a_i - \sum_{i=1}^n ib_i = 0.$$

This follows either by summing (M2), or by the charge relation in (M1). All spherical roots are the ambient simple roots. All colors have exponent $1/2$, and $W_X = S_n \times S_{n+1}$.

Order the coordinates as

$$v_1 > u_1 > v_2 > u_2 > \dots > v_n > u_n > v_{n+1}. \quad (\mathrm{M3})$$

The full virtual multiset is $\{\pm(u_i - v_j)\}_{i,j}$, all with exponent $1/2$. Choose the sign of each weight according to (M3). Every chosen weight is a sum of adjacent differences (M2). The functional with value 1 on every a_i, b_i is well defined, since the two sides of their relation have equal coefficient sum. It is strictly positive on every generator, so none of the opposite signs lies in the color cone. Thus the signed orbits have exactly the claimed positive half. The simple reflection swapping u_i, u_{i+1} crosses exactly b_i, a_{i+1} ; the one swapping v_i, v_{i+1} crosses a_i, b_i . These are the two colors of the corresponding simple spherical root. This proves Statement 7.1.5.

There is no Arthur factor: $e = 0$ and the dual Lie algebra is the full ambient dual Lie algebra, so the centralizer quotient is zero. The even-sphere representation is

$$S_X = (\mathrm{Std}_n^* \otimes \mathrm{Std}_{n+1} \oplus \mathrm{Std}_n \otimes \mathrm{Std}_{n+1}^*)[1].$$

Its weights and positive half are exactly the ones just checked. The precise finite central form is determined by (M1), rather than by the Lie-algebra notation in this display.

The literal simply connected pair is nevertheless outside the manuscript's multiplicity-one hypothesis. Its spherical weight lattice has full rank and lies in the ambient root lattice: H contains the full center of G . To see the latter, for $(\zeta I_n, \eta I_{n+1})$ choose $u = \zeta I_n$ and $\lambda = \eta/\zeta$ in the displayed embedding; then $\lambda^{-n} = \eta$. Since the ambient root lattice is a proper sublattice of the weight lattice, the generic Borel stabilizer has a nontrivial split finite diagonalizable component. Its first Galois cohomology is nontrivial over a nonarchimedean local field, and the generic multiplicity formula gives multiplicity greater than one. Thus this is a geometric and Euler-factor verification, not an assertion that the hypotheses of Theorem 4.5.8 hold for this pair.

17.2 Case 9 in degrees $N \geq 4$

Write $N = m + 2$ and $b = n + 1$, so the ambient group is $\mathrm{SL}_N \times \mathrm{Sp}_{2b}$. Initially take $N \geq 5$ and $b \geq 2$, and omit the central torus in H . Avdeev's six free monoid generators are

$$\lambda_1 = \omega_{N-2}, \quad \lambda_2 = \varphi_2, \quad \lambda_3 = \omega_{N-1} + \varphi_1, \quad \lambda_4 = \omega_1 + \omega_{N-1}, \quad \lambda_5 = \omega_1 + \varphi_1, \quad \lambda_6 = \omega_2.$$

Let $c_1, \dots, c_6$ be the dual color valuations. The color cone is the nonnegative orthant in this basis. Put

$$a = \alpha_1, \quad d = \alpha_2 + \dots + \alpha_{N-2}, \quad a' = \alpha_{N-1}, \quad t = \beta_1, \quad h = \beta_1 + 2\beta_2 + \dots + 2\beta_{b-1} + \beta_b.$$

Their coefficients in the λ basis and their dual coroots are

	1	2	3	4	5	6	coroot
a	0	0	-1	1	1	-1	$c_4 + c_5$
d	1	0	0	-1	0	1	$c_1 + c_6$
a'	-1	0	1	1	-1	0	$c_3 + c_4$
t	0	-1	1	-1	1	0	$c_3 + c_5$
h	0	1	0	0	0	0	$2c_2 + c_3 + c_5$

(M4)

Thus a reflection is computed by $s_\gamma(c) = c - \langle c, \gamma \rangle \check{\gamma}$. The dual semisimple system is $A_3 \times A_1 \times A_1$.

Here is the complete positive multiset, separated by its Euler exponent:

r	positive weights
$\frac{1}{2}$	$c_3, c_4, c_5, c_3 + c_4 + c_5, c_1 + c_4 + c_6, c_1 + c_3 + c_4 + c_5 + c_6$
$\frac{N-3}{2}$	$c_1, c_6, c_1 + c_3 + c_4, c_6 + c_4 + c_5$
$b-1$	$c_2, c_2 + c_3 + c_5$

(M5)

All are nonnegative combinations of colors. Direct substitution in (M4) shows that the union of (M5) and its negative is reflection-stable, and that every row is the orbit multiset generated by its colors. The functional $\ell(c_i) = 1$ separates its two halves. The complete crossing table is

a	c_4, c_5	$(r = 1/2)$
d	c_1, c_6	$(r = (N-3)/2)$
a'	c_3, c_4	$(r = 1/2)$
t	c_3, c_5	$(r = 1/2)$
h	$c_2, c_2 + c_3 + c_5$	$(r = b-1).$

(M6)

This is a finite algebraic certificate valid for every N, b in the stated range; no inference from numerical small-rank tests is involved. The last pair has equal exponents because the height of h is $2b-2$.

For an independent exponent check, $\rho_{P(X)}$ on the four outer SL_N coordinates has coefficients

$$((N-1)/2, (N-3)/2, -(N-3)/2, -(N-1)/2),$$

is zero on the middle block, and on the first two symplectic coordinates equals $be_1 + (b-1)e_2$. Its values on the colors are $((N-3)/2, b-1, 1/2, 1/2, 1/2, (N-3)/2)$, proving all entries of (M5).

The Arthur decomposition of the special-linear standard representation is $\mathbf{4} \oplus V_{N-5}$. The middle summand has dimension $N-4$. The cross Hom spaces give the standard $\mathbf{4}$ and its dual in degree $N-3$. The core even-sphere module is $\bigwedge^2 \mathbf{4} \otimes \mathbf{2}_t$ in degree 1. The symplectic arm contributes $\mathbf{2}_t \otimes \mathbf{2}_h$ in degree $2b-2$ by the same orthogonal Arthur-block calculation as in cases 5-7. Accordingly, up to zero weights,

$$V_X = (\bigwedge^2 \mathbf{4} \otimes \mathbf{2}_t)[1] \oplus (\mathbf{4} \otimes \chi \oplus \mathbf{4}^* \otimes \chi^{-1})[N-3] \oplus (\mathbf{2}_t \otimes \mathbf{2}_h)[2b-2]. \quad (\text{M7})$$

The notation χ records the extra central character; it is suppressed when the central torus is included in H . The actual character lattice determines its normalization. The four positive weights of the middle row in (M5) are precisely the maps from the first two outer coordinates to the middle block and from the middle block to the last two coordinates. The positive symplectic-arm weights are its first and second coordinate weights. Thus the ambient positive-root definition of the quotient agrees with (M5). The six positive core weights are generated by c_3, c_4, c_5 and positive dual simple roots (in particular $c_1 + c_6$), as required for S_X^+ .

Adjoining the torus in H changes the six generators into extended weights with charges $(2, 0, 1, 0, -1, -2)$. On the dual space one therefore imposes

$$2c_1 + c_3 - c_5 - 2c_6 = 0. \quad (\text{M8})$$

This direction is Weyl-fixed by (M4). Since $\ell(c_i) = 1$ annihilates it, the strictly positive functional descends, all positive expressions in (M5) remain positive, and none of their negatives becomes positive. The reflection and crossing tables descend with multiplicities. This proves both versions of case 9 for $N \geq 5$.

For $N = 4$ only the version with the torus is spherical. The same extended weight and reflection calculation applies: $d = \alpha_2$, the exponent $(N - 3)/2$ is $1/2$, and the two standard **4** modules in (M7) now belong to S_X , since the middle Arthur block is absent. For $b = 1$ delete λ_2, c_2, h and the last row of (M5); all other expressions and all remaining crossings stay valid. This deletion applies also to the $N = 3$ calculation below.

17.3 The $N = 3$ endpoint of case 9

Keep the torus in H . Avdeev's generator λ_4 is absent; its former value is $\lambda_1 + \lambda_6$. Use colors $(c_1, c_2, c_3, c_5, c_6)$ and impose the same charge relation (M8). The four spherical roots and coroots are

$$\begin{array}{c|c|c} \alpha_1 & \lambda_1 - \lambda_3 + \lambda_5 & c_1 + c_5 \\ \alpha_2 & \lambda_3 - \lambda_5 + \lambda_6 & c_3 + c_6 \\ \beta_1 & -\lambda_1 - \lambda_2 + \lambda_3 + \lambda_5 - \lambda_6 & c_3 + c_5 \\ h & \lambda_2 & 2c_2 + c_3 + c_5. \end{array}$$

The positive degree-one weights are

$$c_1, c_3, c_5, c_6, c_3 + c_5 + c_6, c_1 + c_3 + c_5,$$

and the arm weights remain $c_2, c_2 + c_3 + c_5$, in degree $2b - 2$. Their reflections cross, respectively, $\{c_1, c_5\}$, $\{c_3, c_6\}$, $\{c_3, c_5\}$ and the arm pair. Substitution in the displayed table verifies closure after adjoining negatives. The functional assigning 1 to every color descends through (M8) and proves sign separation, just as before. The core representation is

$$(\mathbf{3} \oplus \mathbf{3}^*) \otimes \mathbf{2}_t[1],$$

and the centralizer quotient is only the symplectic-arm module, up to zero weights. This proves the endpoint.

A necessary correction to the classification statement. The version of case 9 without $\mathbb{G}_m$ requires $N = m + 2 \geq 5$, as in Avdeev's Table 1, row 4. At $N = 4$, the two distinct irreducible color functions corresponding to λ_1 and λ_6 have the same highest weight ω_2 after forgetting the torus character, so the coordinate representation is not multiplicity free. At $N = 3$ the distinct irreducible generators give the weight relation $\lambda_1 + \lambda_3 = \lambda_6 + \lambda_5$, again contradicting the free weight monoid required for connected semisimple spherical H . Thus these are nonspherical entries if the parameter restriction is omitted; they cannot be checked as cases of Theorem 4.5.8.

The kernel of the printed case-9 parametrization. There is a second, independent correction to Theorem 1.1.28. For $m \geq 1$ the displayed map with the torus is not injective in general:

$$\ker((u, v, w, \lambda) \mapsto ((\lambda^m u) \oplus (\lambda^{-2} v), u \oplus w)) = \{(I_2, \lambda^2 I_m, I_{2n}, \lambda) : \lambda^m = 1\}.$$

Indeed the second component forces $u = I_2, w = I_{2n}$; the first then forces $\lambda^m = 1$ and $v = \lambda^2 I_m$, whose determinant is one automatically. The subgroup is the image, equivalently the quotient of the displayed product by this μ_m . In particular it cannot be called an embedding of that product when $m \geq 2$. This is why extended-character normalizations must use the actual image. For $m = 0$ the whole torus is killed and the image reduces to case 4(a), so that value is a duplicate after taking the image, rather than a new case with a torus factor. The endpoint $n = 0$ is legitimate: interpret Sp_0 as the trivial group and retain the Sp_2 ambient factor. For the version without a torus, $m = 1, 2$ fail sphericity as explained above, while $m = 0$ again reduces to case 4(a).

18 The even-color monoid really gives the asserted positive part

There are two different definitions of positivity to compare. The cone used to check Sakellaridis's Statement 7.1.5 is generated by *all* actual color valuations. In contrast, the monoid in the manuscript's Conjecture 4.5.2, used in Definition 4.5.6, is generated by the *even-sphere* color valuations and the positive dual simple roots. The equality of these positive parts does not follow just by matching the representations. We prove it here, including integral monoid membership rather than merely real cone membership.

Let B be the multiset of weights of S_X , let E be its even-color seed multiset (including the formal even-sphere colors), and write

$$M_E = \mathbb{N}E + \sum_i \mathbb{N}\beta_i, \quad P = B \cap C,$$

where β_i are the dual simple roots and C is the all-color cone. Membership in a monoid is a condition on the weight, so all occurrences of a weight are retained. The case calculations prove $B = P \sqcup (-P)$.

A finite-orbit descent lemma. Suppose B is a finite union of integral Weyl-weight orbits, $E \subset P$, and the following properties hold:

1. Every orbit in B contains a member of E which is crossed by a simple reflection (sent from P to $-P$).
2. For every simple reflection s_i , every crossed weight λ belongs to E and satisfies $\langle \lambda, \beta_i^\vee \rangle > 0$.
3. Every β_i belongs to C .

Then

$$B \cap M_E = P. \tag{SP1}$$

In fact every $\lambda \in P$ is a *single* seed $c \in E$ plus a nonnegative integral sum of the β_i .

To prove this, lower λ by simple reflections with positive coroot pairing whenever the reflected weight remains in P . A strictly dominant linear functional decreases at every such step, so this process terminates because B is finite. If the terminal weight has a positive coroot pairing, its corresponding lowering leaves P , so condition 2 identifies this terminal weight as a seed. If instead it is antidominant, every weight of its Weyl orbit can be reached from it by a sequence of raising simple reflections. A raising reflection cannot leave P , by the positive-pairing clause of condition 2. Thus the whole orbit would lie in P , contradicting condition 1. The terminal weight is consequently

a seed in all cases. Reversing the lowering steps gives the claimed expression, since all coroot pairings are integral. Conversely $M_E \subset C$ by condition 3, which proves (SP1).

The elementary Weyl-orbit assertion just used also holds with central characters present: take the unique antidominant weight in each orbit, choose a minimal representative modulo its stabilizer, and reverse a reduced sequence taking a weight to the antidominant chamber. The Weyl group fixes the central component throughout.

Why its hypotheses have been checked in every retained case. All the displayed S_X modules have weight sets equal to the Weyl orbits of their even-color seeds. These are minuscule modules, tensor products of such modules for different factors, or sums of these modules. No undisplayed interior weights are being discarded. Restrict the complete reflection-crossing tables in the preceding case calculations to their degree-one rows. The crossed weights are exactly even-color seeds, each with positive coroot pairing. Every seed in the table below is crossed. This proves conditions 1 and 2. For condition 3, the formulas in those same tables express each dual simple root as an actual color, twice an even color, or a sum of the two associated positive virtual colors. The latter virtual colors were already proved to lie in the all-color cone. Alternatively, for any displayed crossing θ at that root,

$$\beta_i = \frac{\theta + (-s_i\theta)}{\langle \theta, \beta_i^\vee \rangle} \in C.$$

Here the denominator is positive, as can be read from the same root and color pairings. Thus no additional version of Statement 7.1.5 is assumed in applying the lemma.

Here is the complete seed list for the nonzero S_X cases in the coverage ledger. Superscripts [2] mean two occurrences, not twice a weight. Coordinates and color labels are those in the respective case calculations. These are precisely the lowering-minimal positive weights before a central-character quotient; the expressions and labels descend through every central quotient used above.

case	even-color seeds	
C2, $p = q$	$e_p^{[2]}$	
C8	$x^{[2]}$	
C9, $p = q$	E_p	
C10, $n = 2p$	$E_p^{[2]}$	
C13	$A = f_2, B = -f_3$	
C14	$E_n^{[2]}$	
C15, $\mathrm{SO}_{2n} \setminus \mathrm{SO}_{2n+1}, n \geq 2$	u	
C15, $(\mathrm{SO}_2 \mathrm{SO}_{2n-2}) \setminus \mathrm{SO}_{2n}$	$((u_1 - u_2)/2)^{[2]}$	
G_2/A_2	t	(SP2)
$E_7/(E_6 \mathbb{G}_m)$	$e_3^{[2]}$	
product 3, 7	c_{12}, c_{13}, c_{23}	
product 4(b)	c_1, c_2, c_3, c_5, c_6	
product 4(c), 4(d)	$\delta_1, \dots, \delta_N$	
product 6	c_0, c_A, c_B, c_*	
product 8	$a_1, b_1, \dots, a_n, b_n$	
product 9, $N \geq 5$	c_3, c_4, c_5	
product 9, $N = 4$ with torus	c_1, c_3, c_4, c_5, c_6	
product 9, $N = 3$ with torus	c_1, c_3, c_5, c_6	

The rank-one torus endpoints have two copies of their fundamental weight; they are already represented by C2 with $p = q = 1$. All retained cases not in (SP2) have $S_X = 0$, so require no even-color positivity check. For a formal even-sphere seed, its valuation is the displayed weight although its degree-one occurrence need not be the actual divisor occurrence in the all-color list.

For the standard-module rows, the assertion can also be read directly from $e_i = e_p + (e_i - e_{i+1}) + \dots + (e_{p-1} - e_p)$; the C14 formula uses only indices of the same parity and the corresponding difference

roots $E_i - E_{i+2}$. For C13 the only extra weights are $f_1 = A + (f_1 - f_2)$ and $-f_4 = B + (f_3 - f_4)$. For the rank-two spin row the only extra positive weight is $(u_1 + u_2)/2 = (u_1 - u_2)/2 + u_2$. For products 3 and 7 the extra weight is $x_1 + x_2 + x_3 = c_{12} + 2x_3$. In products 4(c), 4(d), and 8 every color is an even color; the explicit nonnegative integral color expansions already provided prove (SP1) directly. We next give equally explicit certificates for the three less immediate product calculations.

The 32-dimensional spin module in product 4(b)

For $a \geq 4$ put

$$\begin{aligned} \beta_1 &= E_1 - E_2, & \beta_2 &= E_2 - E_3, & \beta_3 &= E_3 - E_4, & \beta_4 &= E_3 + E_4, \\ \beta_5 &= F_1 - F_2, & \beta_6 &= F_2. \end{aligned}$$

A string of six signs below denotes the half of that sign vector in coordinates $(E_1, E_2, E_3, E_4, F_1, F_2)$. This is the complete list of the sixteen positive weights, with integral certificates:

$$\begin{array}{ll|l} - + - + + + & c_2 & - + + - + - & c_3 \\ - + + - + + & c_2 + \beta_3 & + - - + + - & c_1 \\ + - - + + + & c_1 + \beta_6 & + - + - - + & c_6 \\ + - + - + - & c_1 + \beta_3 & + - + - + + & c_1 + \beta_3 + \beta_6 \\ + + - - - - & c_5 & + + - - - + & c_5 + \beta_6 \\ + + - - + - & c_1 + \beta_2 + \beta_3 & + + - - + + & c_1 + \beta_2 + \beta_3 + \beta_6 \\ + + + + - - & c_5 + \beta_4 & + + + + - + & c_5 + \beta_4 + \beta_6 \\ + + + + + - & c_1 + \beta_2 + \beta_3 + \beta_4 & + + + + + + & c_1 + \beta_2 + \beta_3 + \beta_4 + \beta_6 \end{array} \quad (\text{SP3})$$

Every equality follows immediately from the color coordinates (P8). In particular the noneven color $c_4 = E_4$ is never used as a seed. The positive root β_4 is allowed in the definition, even though its reflection has no crossings on S_X .

At $a = 3$, delete E_4 and c_4 . Projection of the even-parity D_4 half-spin sign vectors is a bijection onto the B_3 spin sign vectors: the deleted sign is determined by the other three. The all-color inequality $t_1 + t_2 + t_3 + t_4 \geq 0$ is redundant on these weights once the other five inequalities hold (its only negative case has the first three signs all minus and violates $t_1 + t_2 \geq 0$). Thus it takes the positive half bijectively to the endpoint's positive half. It maps β_3 and β_4 to the positive simple root E_3 . Projecting every equality in (SP3) gives sixteen integral certificates for the $B_3 \times B_2$ endpoint as well.

The 16-dimensional core in product 6

Use half-sign coordinates (t_1, t_2, u, v) as in (P4), and put

$$b_1 = (1, -1, 0, 0), \quad b_2 = (0, 1, 0, 0), \quad b_3 = (0, 0, 1, 0), \quad b_4 = (0, 0, 0, 1).$$

These are the four positive simple roots; the notation here gives their actual coordinates, whereas each sign string denotes half its entries. The eight positive weights have the certificates

$$\begin{array}{ll|l} - + + + & c_* & + + - - & c_0 \\ + - + - & c_A & + - - + & c_B \\ + - + + & c_A + b_4 & + + + - & c_0 + b_3 \\ + + - + & c_0 + b_4 & + + + + & c_0 + b_3 + b_4. \end{array} \quad (\text{SP4})$$

The tail colors do not occur in these expressions. Thus attaching or deleting either tail leaves the even-color monoid argument valid.

The mixed family in product 9

For $N \geq 5$ the degree-one positive weights from (M5) are generated integrally by c_3, c_4, c_5 and the dual simple roots, because

$$\begin{aligned} c_3 + c_4 + c_5 &= c_3 + \check{a}, & \check{a} &= c_4 + c_5, \\ c_1 + c_4 + c_6 &= c_4 + \check{d}, & \check{d} &= c_1 + c_6, \\ c_1 + c_3 + c_4 + c_5 + c_6 &= c_3 + \check{a} + \check{d}. \end{aligned} \tag{SP5}$$

The noneven colors c_1, c_6 need not belong separately to the even-color monoid; their sum is the permitted positive root $\check{d}$. At $N = 4$ with the torus, c_1, c_6 themselves become even seeds, and the two remaining newly degree-one positive weights are

$$c_1 + c_3 + c_4 = c_1 + \check{a}', \quad c_6 + c_4 + c_5 = c_6 + \check{a}.$$

At $N = 3$ with the torus, c_4 is absent. Besides the four seeds c_1, c_3, c_5, c_6 , the positive weights are

$$c_3 + c_5 + c_6 = c_5 + \check{\alpha}_2, \quad c_1 + c_3 + c_5 = c_1 + \check{t},$$

where $\check{\alpha}_2 = c_3 + c_6$ and $\check{t} = c_3 + c_5$. All these equations descend through $2c_1 + c_3 - c_5 - 2c_6 = 0$ when the stabilizer torus is included. The positive functional already used annihilates this relation and remains strictly positive on every color, so no opposite weight enters the monoid after descent. Deleting the arm at $b = 1$ changes none of these equations.

It follows that the degree-one part used in every displayed Euler factor is exactly S_X^+ as defined in the manuscript, with all multiplicities and integral lattice conditions retained. Together with the separate ambient-positive-root checks for the centralizer quotient, this completes the positivity comparison in Definition 4.5.6.

19 Frobenius signs and the final scope check

For split G , the spherical character lattice has trivial Galois action, but the colors and the formal differences of terminal orbits need not. Let η denote an unramified quadratic character, with $\varepsilon = \eta(\varpi) \in \{1, -1\}$. A pair of identical standard modules exchanged by Frobenius is $R \otimes (\mathbf{1} \oplus \eta)$; its two eigenvalues are $1, \varepsilon$, not two copies of 1. For a single formal even-sphere color, the difference of the two terminal orbits transforms by η . These statements follow directly from the permutation action on two basis vectors and their difference.

Family with a potentially nontrivial quadratic action	Effect on the nonzero degree-one weights
C2, equal general-linear blocks	$\text{Std}_{2p} \otimes (\mathbf{1} \oplus \eta)$
C8, $\mathbb{G}_m \text{Sp}_{2n-2} \subset \text{Sp}_{2n}$	$\mathbf{2}_x \otimes (\mathbf{1} \oplus \eta)$
C9, equal symplectic blocks	$\text{Std}_{2p} \otimes \eta$
C10, $\text{GL}_{2p} \subset \text{SO}_{4p}$	$\text{Std}_{2p} \otimes (\mathbf{1} \oplus \eta)$
C14, $\text{GL}_n \subset \text{SO}_{2n+1}$	$W_{\text{parity } n} \otimes (\mathbf{1} \oplus \eta)$
C15, even sphere	$\mathbf{2} \otimes \eta$; at the torus endpoint, $\mathbf{2} \otimes (\mathbf{1} \oplus \eta)$
C15, rank-two even orthogonal blocks	$\text{Spin}_5 \otimes (\mathbf{1} \oplus \eta)$
G_2/A_2	$\mathbf{2} \otimes \eta$
$E_7/(E_6 \mathbb{G}_m)$	$\text{Std}_6 \otimes (\mathbf{1} \oplus \eta)$

Here is a case-by-case justification that the table accounts for the possible nonzero Euler signs in the requested list. Equivariant forms of a fixed homogeneous variety are classified by descent through $\text{Aut}_G(X) = N_G(H)/H$. In the first, third and fourth rows with equal blocks, the normalizer involution exchanges the two equal blocks or the two complementary isotropic spaces.

In C8 it is the Weyl normalizer of the complementary rank-one torus. In C14 the interchange of the isotropic spaces is combined with the appropriate sign on the last orthogonal line to have determinant one. These involutions exchange the two equal-valuation type-T colors, except for the equal symplectic blocks, where they exchange the two terminal orbits defining the single formal even-sphere color. The sphere and rank-two orthogonal actions were computed directly in their sections. The exceptional actions are recorded below and, for G_2 , checked in the functional equation itself. The separate normalizer rows of [BP15, cases 3–4, 10–11, 16–17, 32–33 and 36–37] provide an independent check of these involutions.

The unequal general-linear and unequal symplectic block pairs have no block-exchange automorphism. The versions with a removed central torus, including C3, C6, C11 and case 9 without $\mathbb{G}_m$, have a split torus as their additional connected equivariant automorphism group; $H^1(k, \mathbb{G}_m) = 1$, so it creates no additional form. Their central weight directions must still be retained in all formulas.

For product cases 3, 4(a), 4(b), 5, 6 and 7, the equivariant automorphism groups come from $Z(G)/(Z(G) \cap H)$; their orders, in that order, are 4, 2, 2, 2, 4, 4 away from the separately identified diagonal endpoints. One checks equality by quotienting the displayed eigencharacter lattice by the normalized spherical-root lattice: the respective character groups are $(\mathbb{Z}/2)^2$, $\mathbb{Z}/2$, $\mathbb{Z}/2$, $\mathbb{Z}/2$, $(\mathbb{Z}/2)^2$, $(\mathbb{Z}/2)^2$. There are no additional loose-root involutions in these systems; their simple type-T colors have distinct valuations, and their arm localizations are not rank-one even spheres. Central automorphisms fix colors and act trivially by ambient adjoint action. The nonzero S_X is irreducible in these product cases and contains a fixed canonical color vector; hence its compatible Frobenius action is the identity. The central twists of the diagonal and Gross–Prasad cases likewise fix their color vectors; the matrix-block description of the stabilizer adds no exchange of the two differently sized ambient factors. Case 8 and the torus-containing case 9 have no additional normalizer involution. At an endpoint, use the normalizer of the identified lower-rank pair.

For every other classical case in the ledger the degree-one module is zero, or its different color valuations force its canonical vectors to be fixed. In particular the two **4** constituents of the $\mathrm{SO}_2 \mathrm{Spin}_7$ case have different color valuations; they are not an exchangeable duplicate pair. Since the ambient pinned dual group is split, the nonzero Lie-quotient Euler weights retain sign $+$. Equivalently, the non-simple type-T roots in these product localizations have color valuations not proportional to half their spherical coroot, so they cannot acquire Sakellaridis’s nonsplit-T sign. Possible changes on purely zero torus weights only change a scalar factor and are irrelevant to (V2).

The cone proofs remain valid with these signs. Where two copies are exchanged, split their multiplicity space into the $+1$ and -1 eigenspaces and attach the same sign to the entire Weyl orbit. Where a formal single color has sign ε , do the same for its standard orbit. Each crossing table is an equality with multiplicity labels, so it continues to be an equality of signed, graded multisets. This proves the signed version of Statement 7.1.5, rather than discarding its signs.

20 Changes recommended in the manuscript

Replace the proof of Theorem 4.5.8 by the reflection reduction and the case certificates, citing the later rank-one scattering result for the exceptional supports. State the integral-model hypotheses already used for the local spherical functions and specify whether the theorem is restricted to the connected classification checked here. Keep (4.5.1) as a genuine hypothesis; it is false for several entries even generically, and can also fail at special unramified parameters when the generic multiplicity is one.

Correct Definition 4.5.6 by (V3), use positive roots rather than positive coroots of the dual group, and describe $\check{\mathfrak{g}}_X^+$ as a torus representation. Include the omitted root-operator clause when quoting the BZSV conjecture. In Definition 4.5.1 retain multiplicities and distinguish a highest-weight recipe from a proved assertion about all weights.

Correct the parameter range and finite kernel in item 9 of Theorem 1.1.28, and specify the

Spin lifts in its orthogonal items. These corrections matter both to sphericity and to the number of rational open Borel orbits. In particular, replacing a simply connected pair by an effective orthogonal or adjoint pair can remove a multiplicity obstruction; it is not an innocuous change to the local theorem’s hypothesis.

Source discrepancies not to copy into the manuscript. In arXiv:0905.4244v3, equation (6.13) prints the even-sphere exponent n where the virtual-color calculation gives $n - 1/2$. Appendix A, row 4, prints standard exponents $1/2, 3/2$ independently of the difference of the two symplectic block ranks and prints exterior-square exponent 1; the verified values are $\delta + 1/2, \delta + 3/2$ and 2. Row 6 likewise prints exterior-square exponent 1 where it is 2. These corrections follow twice, from $\langle \check{v}_D, \rho_{P(X)} \rangle$ and from the Arthur highest weight plus two, in the calculations above. The even-sphere correction also agrees with the later rank-one table. They are discrepancies in that explicitly inspected version, not counterexamples to the corrected cocycle formula.

References

- [MS] The author’s supplied manuscript, *Spherical functions on spherical varieties for unramified groups*, Theorem 1.1.28 and §4.5, especially Definitions 4.5.1–4.5.6 and Theorem 4.5.8. Page numbers in this note refer to the supplied PDF’s printed numbering (the local theorem is on p. 152).
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- [Sak08] Y. Sakellaridis, *On the unramified spectrum of spherical varieties over p -adic fields*, Compos. Math. **144** (2008), 978–1016. [arXiv:math/0606130](#).
- [Sak25] Y. Sakellaridis, *Transfer operators and Hankel transforms: horospherical limits and quantization*, in *Symmetry in Geometry and Analysis*, Vol. 1, Progress in Mathematics 357 (2025), 435–495. [arXiv:2407.19500](#); Theorem 2.4.2 and its Radon-inversion proof.
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